

FLUOR DANIEL

CONCRETE COLUMN DESIGN

PURPOSE

This practice defines and illustrates the design procedures for rectangular and circular concrete columns subjected to uniaxial or biaxial bending and axial compression or tension. In addition, an engineer with computer programming expertise will be able to convert the step-by-step algorithm presented here into a useful computer program.

SCOPE

Emphasis

This practice focuses on biaxial bending of rectangular sections, with uniaxial bending as a simple, specialized case. By virtue of symmetry, biaxial bending of circular columns can be reduced to uniaxial bending.

Basis

Bresler's Load Contour and Load Reciprocal methods are the basis of strength design for biaxial bending.

Summary

- General discussion of compatibility and equilibrium requirements.
- Overview of the developments leading to the procedures for biaxial bending described herein.
- Discussion of Load Contour Method for obtaining an appropriate equivalent uniaxial moment for rectangular columns.
- Choosing a trial section for analysis based on the equivalent uniaxial moment.
- Using Load Contour and Reciprocal Methods to analyze the trial section for biaxial bending and compression.
- Extending the procedures to biaxial bending and tension.
- Procedures for slender columns.
- Design examples.

APPLICATION

Use Of Practice

The user of this practice need not read the in-depth theory and discussion presented herein unless it is intended to develop a comprehensive spreadsheet program. Instead, the applicable Sample Design may be reviewed. Attachment 15: General Notes for the Sample Designs, immediately preceding the sample calculations should be read first. However, design engineers are encouraged to understand the philosophy behind the procedures to enhance their judgement during design.

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Summary Of Design Procedure

Based on the appropriate Sample Designs, use the following procedure:

- Choose a trial section.
- If required, check for slenderness and refine the trial section.
- Analyze the trial section.
- Ensure that final details satisfy all requirements of the code.

DEFINITIONS

Subscripts

- n - nominal
- u - applied factored
- x - about x-axis or along x-axis as noted
- y - about y-axis or along y axis as noted
- o - uniaxial with $e_x = 0$ or $e_y = 0$
or axial with $e_x = e_y = 0$

Notations

Dimensions

- b - width of section and is the smaller dimension
- h - depth of section and is the larger dimension
- X-axis - an axis passing through the plastic centroid and parallel to side b
- Y-axis - an axis passing through the plastic centroid and parallel to side h ($h > b$)
- e_x - eccentricity measured along X-axis, $= M_{oy}/P_u = M_{oy}/P_n$
- e_y - eccentricity measured along Y-axis, $= M_{ux}/P_u = M_{ux}/P_n$
- e_{ox} - eccentricity measured along X-axis, $= M_{uoy}/P_u = M_{uoy}/P_n$
- e_{oy} - eccentricity measured along Y-axis, $= M_{uox}/P_u = M_{uox}/P_n$

Factored Loads

- P_u - applied factored compressive load
- M_{ux} - applied factored moment about the X-axis

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Load Capacities

- M_{uy} - applied factored moment about the Y-axis
- M_{uox} - equivalent uniaxial factored moment about the X-axis
- M_{uoy} - equivalent uniaxial factored moment about the Y-axis
- P_n - nominal compressive load capacity associated with both $e_x > 0$ and $e_y > 0$
- M_{nx} - nominal moment capacity about X-axis associated with P_n and M_{uy}
- M_{ny} - nominal moment capacity about Y-axis associated with P_n and M_{ux}
- M_{nox} - nominal uniaxial moment capacity about X-axis associated with P_n only, such that $P_n = P_u/\phi$
- M_{noy} - nominal uniaxial moment capacity about Y-axis associated with P_n only, such that $P_n = P_u/\phi$

Additional Notations (For Reciprocal Load)

- P_{no} - nominal axial compressive load capacity at $e_x = e_y = 0$
- P_{nox} - nominal axial compressive load capacity at $e_y > 0$, $e_x = 0$
- P_{noy} - nominal axial compressive load capacity at $e_y = 0$, $e_x > 0$

Other notations are set out in the Figures included in Attachments 1 through 14.

CODE

Unless noted otherwise, code sections and equations refer to ACI 318-89.

GENERAL

Fundamentally, there is no difference between uniaxial bending and biaxial bending of concrete columns. Both are based on conditions of strain compatibility and equilibrium of forces. Biaxial bending, however, requires more tedious arithmetic and many more iterations for a solution.

For a given section under loads P_u and M_u , uniaxial bending requires merely the determination of capacities P_n and M_n so that $e = M_u/P_u = M_n/P_n$ and $P_u < P_n$. Using a straight line strain distribution and the Whitney stress block as shown in Attachment 03 makes this a relatively simple procedure.

Skew bending, however, requires the determination of the vector (P_n, M_{nx}, M_{ny}) such that applied eccentricities $e_y = M_{ux}/P_u = M_{nx}/P_n$ and $e_x = M_{uy}/P_u = M_{ny}/P_n$ act simultaneously, and $P_u < P_n$.

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For any orientation θ of the neutral axis, determination of (P_n, M_{nx}, M_{ny}) can be tedious (refer to Attachment 04). In addition, various orientations, θ , have to be tried until the moment capacities M_{nx} and M_{ny} are such that $e_y = M_{nx}/P_n$ and $e_x = M_{ny}/P_n$, where e_y and e_x are the applied eccentricities.

An alternative procedure is to study the failure surfaces shown in Attachment 07 and presented in the developments by Bresler, Meek, and Parme discussed below.

Compatibility

Equations of compatibility may be expressed as follows:
(Refer to Attachments 01, 02, 03 and 04.)

Equation 1

$$e_c = 0.003 \quad \text{ACI Section 10.2.3}$$

$$e'_s = \left(\frac{c - d'_s}{c} \right) e_c$$

$$e''_s = \left(\frac{c - d''_s}{c} \right) e_c$$

$$e_s = \left(\frac{c - d}{c} \right) e_c$$

Equation 2

$$f'_s = e'_s E_s \text{ but not greater than } f_y$$

$$f''_s = e''_s E_s \text{ but not greater than } f_y$$

$$f_s = e_s E_s \text{ but not greater than } f_y$$

Equilibrium

Equations of equilibrium may be expressed as follows:
(Refer to Attachments 1, 2, 4 and 5.)

Equation 3

$$C' = f'_s A'_s$$

$$C'' = f''_s A''_s$$

$$T = f_s A$$

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Equation 4a

For bending about X-axis:

$$c = 0.85 f'_c b a$$

$$\text{where } a = \beta_1 c$$

$$\beta_1 = 0.85 - \frac{f'_c - 4000}{1000} 0.05 \text{ but not less than } 0.65$$

(ACI Section 10.2.7.3)

For bending about Y-axis:

Equation 4b

$$C = 0.85 f'_c h a$$

Note!!! The equation $C = 0.85 f'_c b a$ ignores the concrete displaced by the compression steel. This is the assumption followed by the ACI Handbook, and this assumption will be followed by this practice so that the spreadsheet calculation may be compared with the ACI Handbook.

However, if it is required to make allowance for the concrete displaced by the compression steel, then replace the compression stress f'_s by $f'_s *$, f''_s by $f''_s *$, etc., where $f'_s * = f'_s - 0.85 f'_c$ and so on.

Equation 5

$$l'_a = d - d'_s$$

$$l''_a = d - d''_s$$

$$l'''_a = d - d'''_s$$

$$l_a = d - \frac{a}{2} = d - \frac{\beta_1 c}{2}$$

Equation 6

$$P + \sum T = \sum C$$

This will solve P.

Taking moment about the tension steel A_s ,

Equation 7

$$Pe' = C' l'_a + C l_a + C'' l''_a - T''' l'''_a$$

This will solve e' .

Then e may be found as shown in Attachment 01, Figure 2.

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Balanced Condition

At balanced condition, the concrete reaches its maximum usable strain at the same time the tension steel A_s yields.

Equation 8

$$e_c = 0.003 \quad \text{ACI Section 10.2.3}$$

$$e_s = \epsilon_y = \frac{f_y}{E_s}$$

$$c_b = \left(\frac{0.003}{0.003 + \epsilon_y} \right) d$$

Compatibility and Equilibrium equations are then used to give load, P_b , and eccentricity, e_b , at balanced condition.

BIAXIAL BENDING DEVELOPMENT

Bresler

For uniaxial bending, compatibility and equilibrium equations discussed in the previous section produce a 2 dimensional interaction diagram shown in Attachments 5 and 6. This 2 dimensional failure curve transforms to a 3 dimensional surface for biaxial bending. Figure 1 in Attachment 07 shows a plot of the vector (P, M_x, M_y) that forms the basis for the load Contour Method; and Figure 2 in Attachment 07 shows $(1/P, e_y, e_x)$ that leads to the Reciprocal Load Method.

Load Reciprocal Method

For this method, Bresler suggested a failure plane (Attachment 07, Figure 2), the equation for which is determined by analytical geometry to be:

Equation 9

$$\frac{1}{P_n} \propto \frac{1}{P'_n} = \frac{1}{P_{nox}} + \frac{1}{P_{noy}} - \frac{1}{P_{no}}$$

where P'_n occurs on the true failure surface and P_n occurs on the failure plane. Each point on the true surface is approximated by a different plane. This procedure is most suitable when P_{nox} and P_{noy} are greater than the balanced axial force P_b for the particular axis, and should only be used when P_n is greater than or equal to $0.1 f'_c h b$.

Load Contour Method

For a constant load, P , the vector (M_x, M_y) gives a 2 dimensional load contour shown in Attachment 08, Figures 1 and 2. The elliptical contour shown in Figure 1 is a good approximation of a somewhat irregular curve. The nondimensional plot of Figure 2 has the important feature that the load contour is virtually symmetrical about the 45 degree axis. Bresler suggested the following unity equation:

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Equation 10a

$$\left(\frac{M_{nx}}{M_{nox}}\right)^{N1} + \left(\frac{M_{ny}}{M_{noy}}\right)^{N2} = 1.0$$

Bressler also assumed $N1 = N2$

$$\therefore \left(\frac{M_{nx}}{M_{nox}}\right)^N + \left(\frac{M_{ny}}{M_{noy}}\right)^N = 1.0$$

Note!!! Bresler used the notation α and β instead of $N1$ and $N2$ respectively. This practice uses $N1$ and $N2$, since the symbol β is used later in a different meaning.

Meek

Meek calculated the theoretical contour line for a particular column, and compared the curve with experimental results. The results are shown in Attachment 09, Figure 1. He suggested the bilinear relationships shown in Attachment 09, Figure 2.

Parme And PCA

Parme et al. extended the Load Contour Method proposed by Bresler as follows:

Equation 10b

$$\left(\frac{M_{nx}}{M_{nox}}\right)^N + \left(\frac{M_{ny}}{M_{noy}}\right)^N = 1.$$

A control point on the curve is defined such that

$$\frac{M_{nx}}{M_{nox}} = \frac{M_{ny}}{M_{noy}} = \beta$$

$$\therefore \beta^N + \beta^N = 1.0$$

$$2\beta^N = 1.0$$

$$\beta^N = 0.5$$

$$N \log \beta = \log 0.5$$

$$\therefore N = \frac{\log 0.5}{\log \beta}$$

$$\therefore \left(\frac{M_{nx}}{M_{nox}}\right)^{\frac{\log 0.5}{\log \beta}} + \left(\frac{M_{ny}}{M_{noy}}\right)^{\frac{\log 0.5}{\log \beta}} = 1.0$$

More importantly, Parme et al. developed a comprehensive set of charts giving the values of β for various reinforcing bar arrangements, steel percentage, f'_c , f_y , and ratio $(P_v/\Phi)/P_{no}$. These charts are shown in Attachments 11 through 14. Nondimensional failure curves for various values of β are shown in Attachment 13.

Note!!! For Attachments 11 through 14, replace P_u with P_v/Φ and P_o with P_{no} .

These charts, together with uniaxial bending diagrams and the unity equation provide the basis for the method most commonly used today.

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EQUIVALENT UNIAXIAL MOMENT

Load Contour Method

PCA adopted Parme's approach, and republished his paper in Advanced Engineering Bulletin No. 18. The ACI Handbook, Volume 2 also adopted this approach.

Equation 11a

$$M_{nox} = \sqrt[N]{M_{nx}^N + \left(\frac{M_{nox}}{M_{noy}} M_{ny}\right)^N}$$

$$\text{Where } N = \frac{\log 0.5}{\log \beta}$$

Equation 11b

$$\frac{M_{nox}}{M_{noy}} \propto \frac{h}{b}$$

$$\therefore M_{nox} \propto \sqrt[N]{M_{nx}^N + \left(\frac{h}{b} M_{ny}\right)^N}$$

For $\beta = 0.5$, this reduces to

Equation 12a

$$M_{nox} = M_{nx} + \frac{h}{b} M_{ny}$$

Equation 12b

$$= M_{nx} + M_{ny} \text{ for } h=b$$

Parme's charts show that β never reaches 0.5, and only approaches that value for light axial loads and high reinforcement ratio. The linear relation, expressed by Equation 12, is therefore quite conservative.

Equation 13a

$$\text{For } \beta = \frac{1}{\sqrt{2}}, \quad N = \frac{\log 0.5}{\log \left(\frac{1}{\sqrt{2}}\right)} = 2$$

$$M_{nox} = \sqrt{M_{nx}^2 + \left(\frac{h}{b} M_{ny}\right)^2}$$

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Equation 13b

$$= \sqrt{M_{nx}^2 + M_{ny}^2} \text{ for } h = b$$

Both Equations 12 and 13 are often used to establish an approximate uniaxial moment for design purpose.

Parme and PCA use the bilinear relation of Meek, shown in Figure 2 of Attachment 09, to establish the equivalent uniaxial moment. This necessitates estimating beforehand whether M_{nx}/M_{nox} is greater than or less than M_{ny}/M_{noy} . In this practice, Equation 11 will be used to establish the equivalent uniaxial moment capacity.

Examination of Parme's charts shows that $\beta \sim 0.65$

for usual values of $q = \frac{A_{st}}{hb} \frac{f_y}{f'_c}$ between 0.1 and 0.3.

Therefore, for $\beta = 0.65$,

$$\text{Equation 11b becomes } M_{nox} = 1.6 \sqrt{M_{nx}^{1.6} + \left(\frac{h}{b} M_{ny}\right)^{1.6}}$$

TRIAL SECTION

Equation 14

A trial section for further analysis may then be established as follows:

Since $M_n \propto M_u$,

$$\therefore M_{uox} = 1.6 \sqrt{M_{ux}^{1.6} + \left(\frac{h}{b} M_{uy}\right)^{1.6}}$$

For economy, use $\frac{h}{b} \propto \frac{M_{ux}}{M_{uy}}$ but not greater than 2/1 unless h/b ratio is given, based on other considerations, for example, Architectural.

$$\text{For pure flexure, } A_s \propto \frac{M_{uox}}{0.08 h (\phi f_y)}$$

For practical section, $A_s \sim 0.015hb$

$$\therefore 0.015 hb \propto \frac{M_{uox}}{0.08 h (\phi f_y)}$$

$$\therefore h^2 b = 83.33 \frac{M_{uox}}{\phi f_y}$$

Choose a trial section based on pure flexure, that is, use $\phi = 0.9$.

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Equation 15

$$\therefore h^2 b = \frac{83.33 M_{uox}}{0.9 f_y} = 93 \frac{M_{uox}}{f_y}$$

Because the ratio h/b is already chosen as discussed above, h can be found, and b can be found.

Equation 16

$$\text{Then } A_s = 2 A_s \approx 0.03 hb$$

This will give a good, reasonable section for further analysis, using either the Load Contour Method or Load Reciprocal Method.

BIAXIAL BENDING AND COMPRESSION

Load Contour Method To Check Trial Section

- Using the trial section, find β , M_{nox} , M_{noy} at load P_u/ϕ

Equation 17

- Find $N = \frac{\log 0.5}{\log \beta}$
- If the section is adequate, then

Equation 18

$$\begin{aligned} \frac{M_{ux}}{\phi} &\leq M_{nx} \\ \frac{M_{uy}}{\phi} &\leq M_{ny} \\ \therefore \left(\frac{\frac{M_{ux}}{\phi}}{M_{nox}} \right)^N + \left(\frac{\frac{M_{uy}}{\phi}}{M_{noy}} \right)^N &\leq 1. \end{aligned}$$

Load Reciprocal Method To Check Trial Section

- Check to ensure that $P_u/\phi \geq 0.1 f'_c hb$. If not, do not use Load Reciprocal Method.
- Find P_{no} , P_{nox} , P_{noy}
- Find P_n using Equation 9
- If trial section is adequate, $P_n \geq P_u/\phi$

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BIAXIAL BENDING AND TENSION

The methods above may be extended to biaxial bending and tension as follows:

- Determine trial section based on the equivalent uniaxial bending moment as discussed above. This gave h , b , and A_{st} .
- Using the trial section, find β , M_{nox} , M_{noy} at load $P_n = 0$.
- Find $N = \frac{\log 0.5}{\log \beta}$
- Check that section is adequate for pure flexure (that is, $P_n = 0$) using Equation 18.

Equation 19

- Increase the reinforcing steel such that $A_{ST} = A_{st} + \frac{T_u}{\phi f}$
- The section with h , b , and A_{ST} , as determined, will be adequate. However, the final section may be checked, preferably by a spreadsheet program, using the equivalent uniaxial moment found above and the applied tension T_u . Invariably it will be found that A_{ST} may be reduced.

SLENDER COLUMNS

Evaluation of slenderness

effects is in accordance with ACI Section 10.11. The moments and both ends of the column must be considered. Also, minimum eccentricity requirements must be satisfied. It should be noted that ACI 318.83 and later editions do not require a minimum eccentricity for short (non-slender) columns. The minimum eccentricity is replaced by a maximum design load.

The object of the design is to find a section (h , b , A_{st}) that satisfies the unity Equation 20 for the magnified moments at each end of the column:

Equation 20

$$\left(\frac{\frac{M_{ex}}{\phi}}{M_{nox}} \right)^N + \left(\frac{\frac{M_{ey}}{\phi}}{M_{noy}} \right)^N \leq 1.0$$

where $M_e = \delta_b M_b + \delta_s M_s$, where δ_b and δ_s are the magnification factors, M_b and M_s are moments due to gravity and lateral loads respectively. However, δ_b and δ_s depend on the section we are trying to find; hence the process is iterative.

The following procedures will reduce the iterative process, if not eliminate it completely.

- Neglecting slenderness, find equivalent uniaxial moments M_{nox} at each end of the column, using Equation 14.

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- Find vector (h, b) for each end of the column, using Equation 15.
- Use larger 'h' and larger 'b' for the dimensions of the section.
- Check if slenderness should be considered using ACI Section 10.11.4.
- If slenderness is a consideration, magnify the larger of the equivalent uniaxial moments at the top and bottom of the column and re-estimate the trial section h , b , and A_s . The magnification of the uniaxial moment at this stage is a judgment call and depends on how much the $k L/r$ ratio exceeds the limits of ACI Section 10.11.4. If in doubt, use $1.5 M_{u_{max}}$ to re-estimate trial section.
- We now have a trial section for analysis. Apply ACI Section 10.11.5 to determine M_{cx} and M_{cy} at each end of the column.
- Check the minimum eccentricity requirements of ACI Section 10.11.5.4. At each end of the column use the greater value of M_{cx} determined from minimum eccentricity or the computed M_{cx} above, and the greater M_{cy} determined from minimum eccentricity or the computed M_{cy} above.
- Check each end of the column for the respective vectors, (M_{cx}, M_{cy}, P_u) using Equation 20 and the procedures outlined in previous section.
- These procedures should give an adequate section. If not, or if better economy is desired, adjust section and repeat calculation.

EXAMPLES

Examples to illustrate the above procedures are included in Attachment 16.

SPREADSHEET PROGRAM

The step-by-step procedures above, together with the examples in Attachment 16, may be used to prepare a spreadsheet program. If such a program is prepared, ensure that Code (ACI 318-89) requirements are met, for example, minimum steel, appropriate values of ϕ , β_1 , and similar considerations.

In addition, the Parme's charts for β (shown in Attachments 11 through 14) may be converted into Lookup Tables that could be stored in a database for retrieval. Or, the value of β may be read from the charts and entered as input data; this latter method, however, reduces the automotive feature of the program.

REFERENCES

Bresler, Boris. Design Criteria for Reinforced Columns Under Axial Load and Biaxial Bending. Journal of the American Concrete Institute. November, 1960.

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Building Code Requirements for Reinforced Concrete (ACI 318-89) and Commentary (ACI 318R-89).

Meek, John L. Ultimate Strength of Columns with Biaxially Eccentric Loads. Journal of the American Concrete Institute. August, 1963.

Parme, Alfred L., Jose M. Nieves, and Albert Gouwens. Capacity of Reinforced Rectangular Columns Subject to Biaxial Bending. Journal of the American Concrete Institute. September, 1966.

ATTACHMENTS

Attachment 01:

Figure 1. Basic Nomenclature

Figure 2. Nomenclature Eccentricity

Attachment 02:

Nomenclature For Reinforcement

Attachment 03:

Stress And Load Distribution For Compatibility And Equilibrium

Attachment 04:

Skew Bending

Attachment 05:

Load-Moment Interaction Diagrams

Attachment 06:

Load-Moment Interaction Diagram Comparisons

Attachment 07:

Figure 1. Load Contours For Constant P_n On Failure Surface S_3

Figure 2. Reciprocal Load Method

Attachment 08:

Figure 1. Load Contour At Plane Of Constant P_n Of Failure Surface S_3

Figure 2. Nondimensional Load Contour At Constant P_n

Attachment 09:

Figure 1. Theoretical Contour Line On Interaction Surface

Figure 2. Bilinear Approximation Of Load Contour Line

Attachment 10:

Nondimensional Failure Curves For Various Values Of β

Attachment 11:

Biaxial Bending Design Constants

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Attachment 12:
Biaxial Bending Design Constants

Attachment 13:
Biaxial Bending Design Constants

Attachment 14:
Biaxial Bending Design Constants

Attachment 15:
General Notes For Sample Design

Attachment 16:
Sample Design 1: Biaxial Bending - Trial Section

Attachment 17:
Sample Design 2: Biaxial Bending And Axial Compression

Attachment 18:
Sample Design 3: Biaxial Bending And Axial Tension

Attachment 19:
Sample Design 4: Biaxial Bending And Axial Compression For Slender Columns

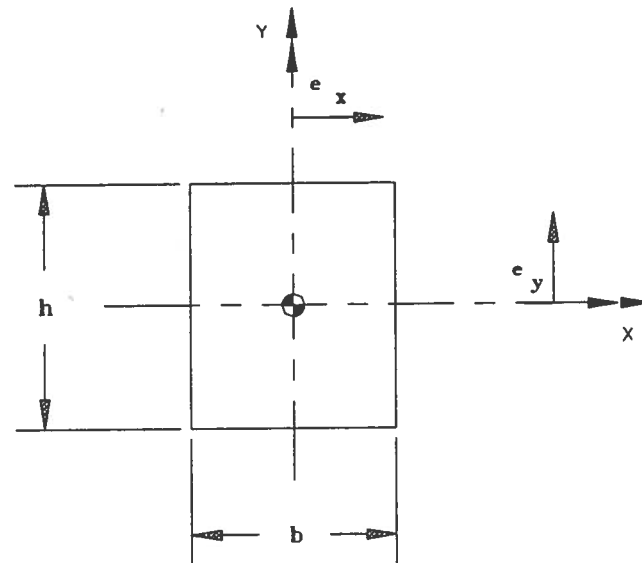
Attachment 20:
Sample Design 5: Uniaxial Bending And Axial Compression

Attachment 21:
Sample Design 6: Circular Spiral Column

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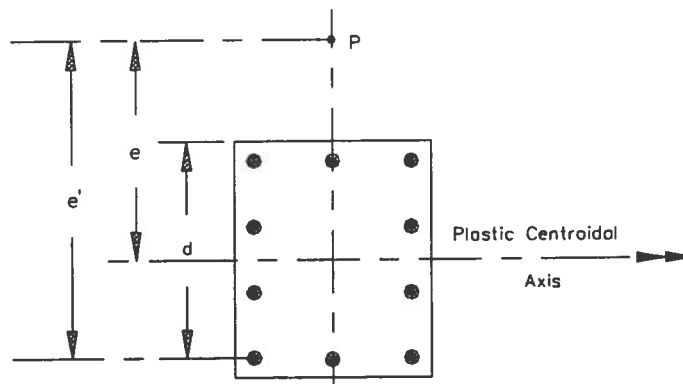
Figure 1. Basic Nomenclature



"h" is greater than "b"

● = Plastic Centroid, which does not necessarily lie at the geometric center

Figure 2. Nomenclature Eccentricity



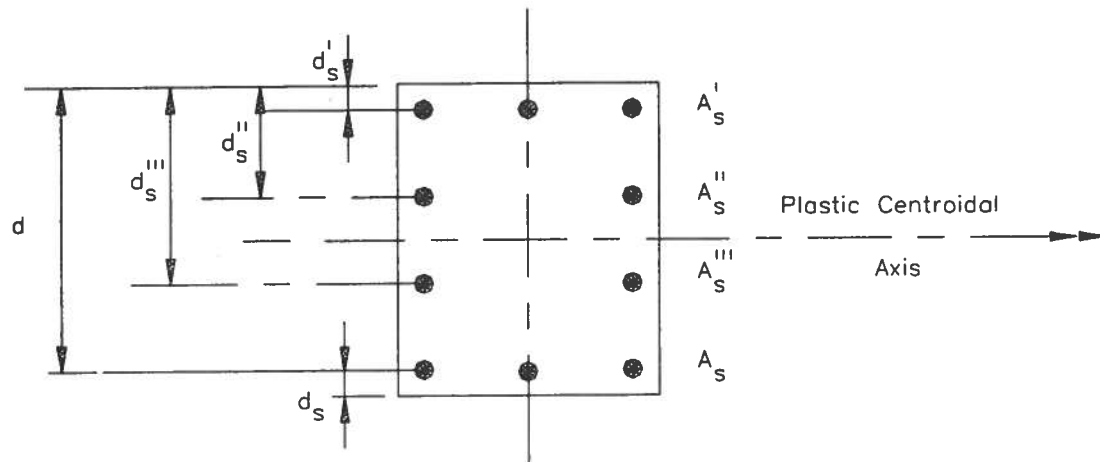
If Plastic Centroid lies at the geometric center

Then: FOR BENDING ABOUT THE X AXIS
 $e = e' - d + (h/2)$

and: FOR BENDING ABOUT THE Y AXIS
 $e = e' - d + (b/2)$

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NOMENCLATURE FOR REINFORCEMENT



A_s^I = Compression Steel

A_s = Tension Steel

A_s^{II} is in compression if depth to neutral axis c is greater than d_s^{II} and in tension if c is less than d_s^{II}

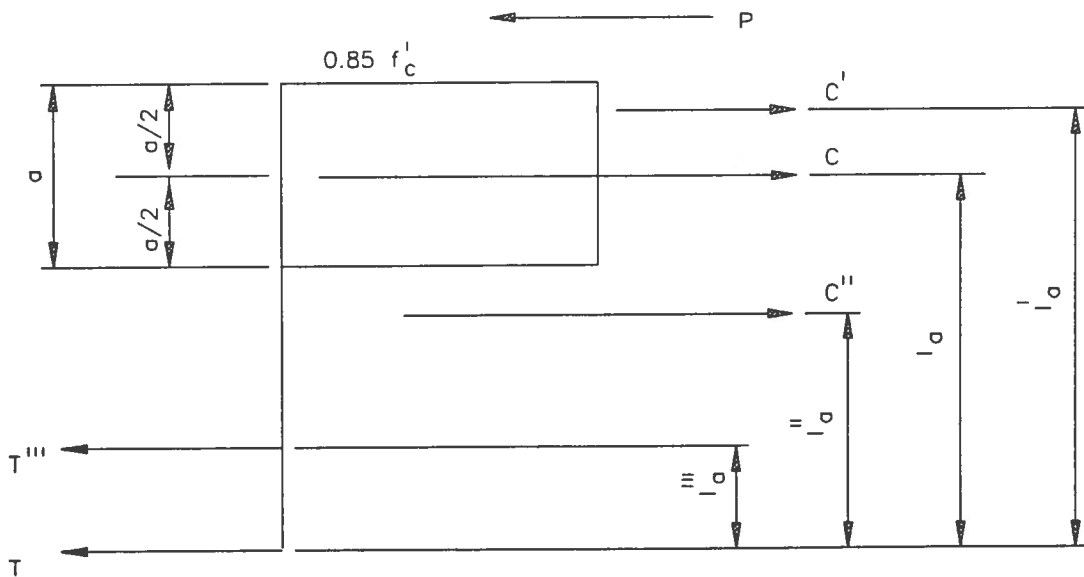
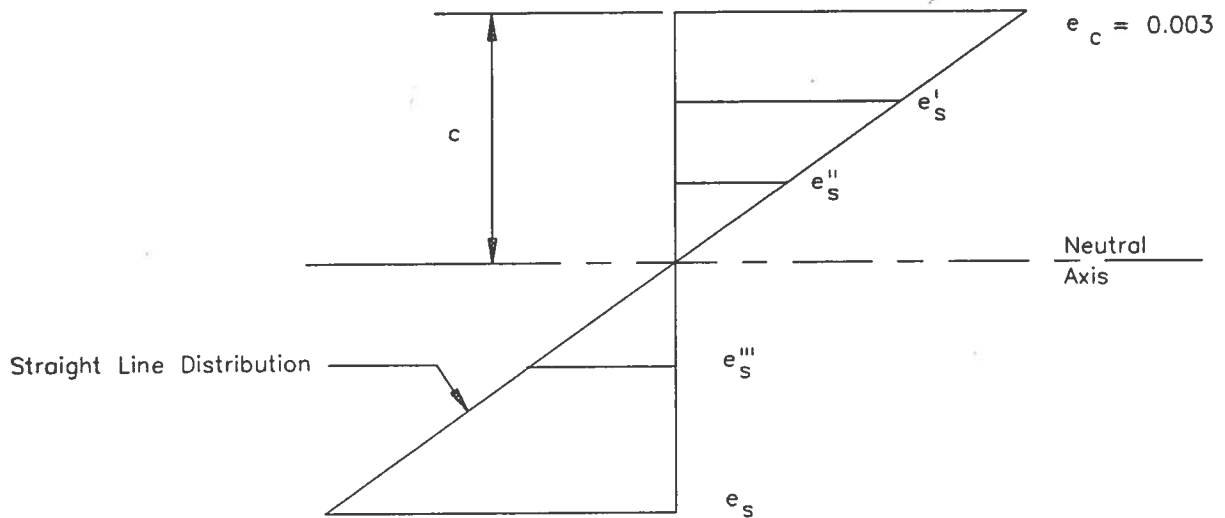
A_s^{III} is in compression if c is greater than d_s^{III} and in tension if c is less than d_s^{III} , and so on

$$A_{st} = A_s^I + A_s^{II} + A_s^{III} + \dots + A_s$$

$$P = \frac{A_{st}}{hb} ; \quad q = \frac{A_{st}}{hb} \times \frac{f_y}{f_c'}$$

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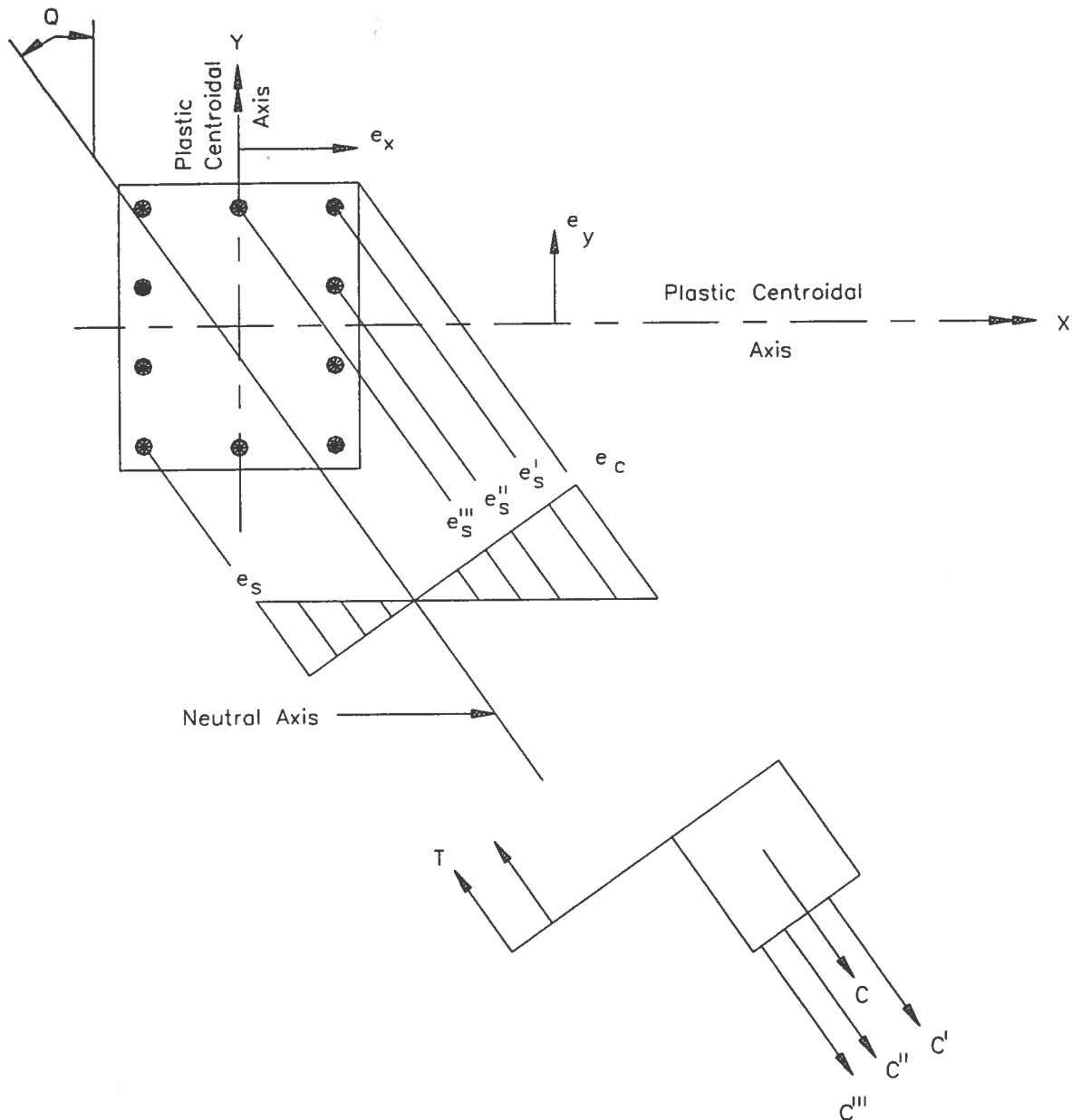
STRESS AND LOAD DISTRIBUTION FOR COMPATIBILITY AND EQUILIBRIUM



$$P + \text{Sum}(T) = \text{Sum}(C)$$

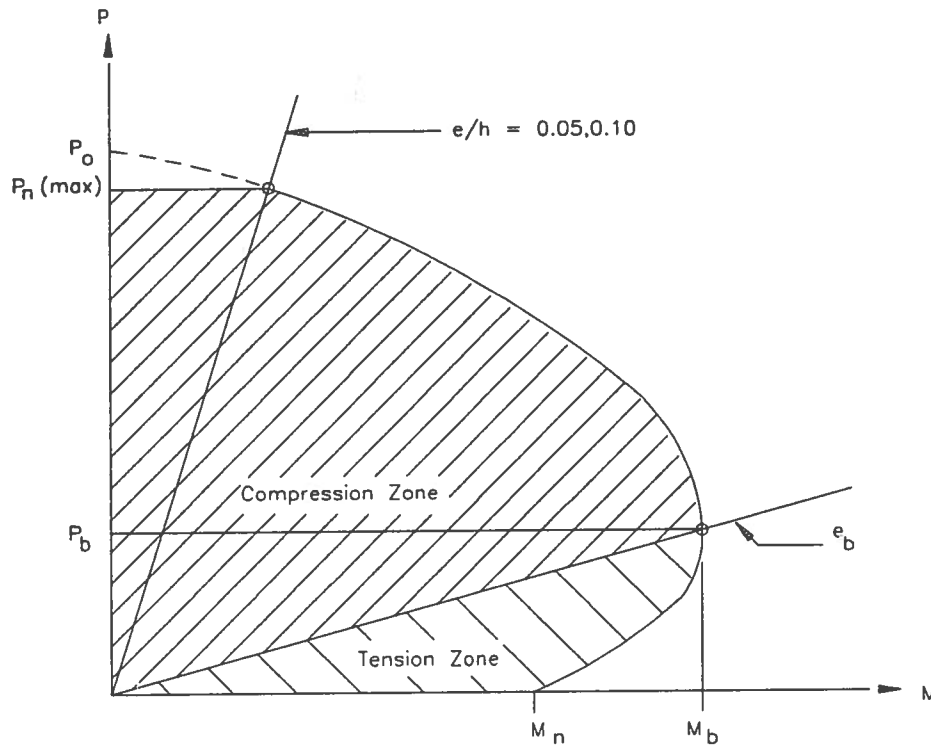
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SKEW BENDING

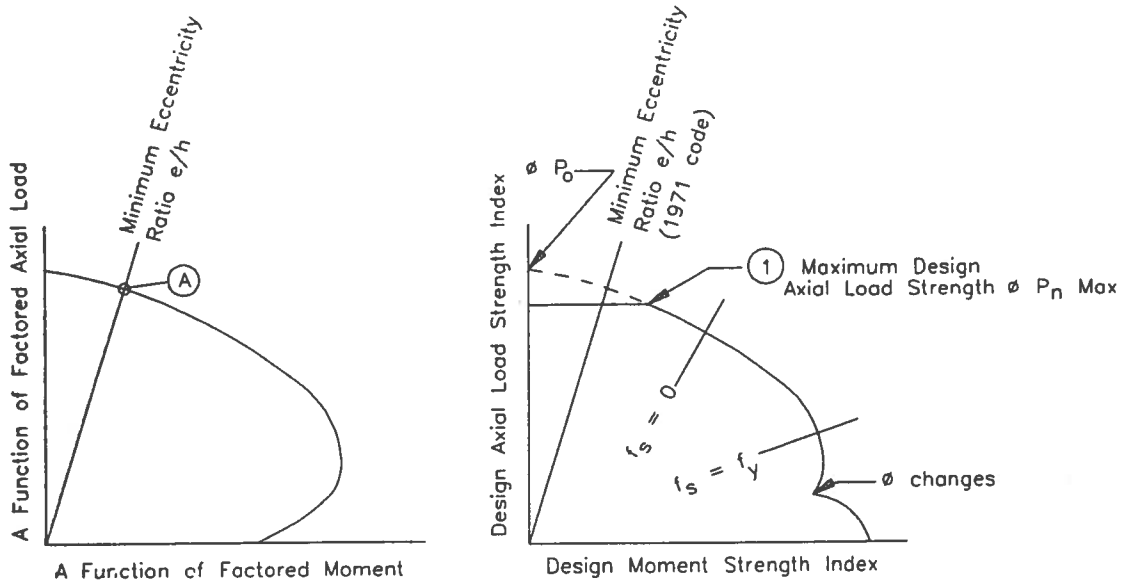


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LOAD - MOMENT INTERACTION DIAGRAM



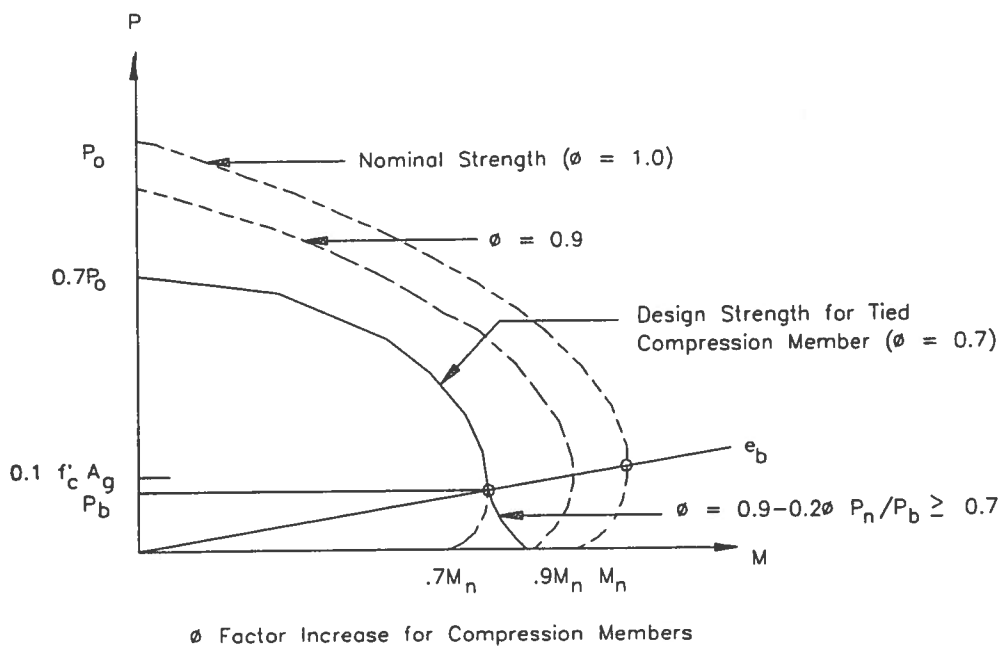
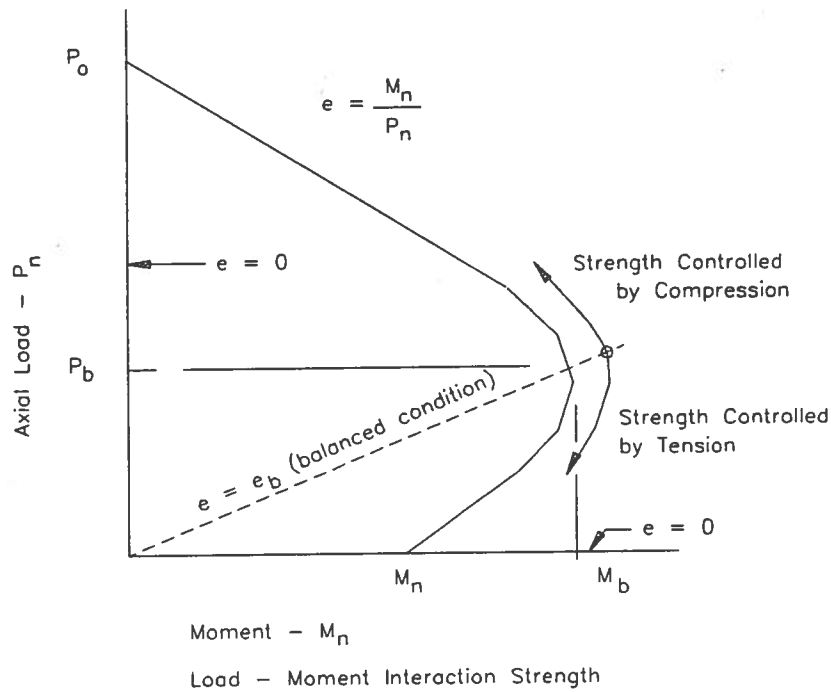
General Form of Load - Moment Interaction Strength



Comparison of interaction diagrams according to 1971 Code (left) and later Codes (right)

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LOAD - MOMENT INTERACTION DIAGRAM COMPARISONS



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Figure 1. Load Contours for Constant P_n on Failure Surface S_3

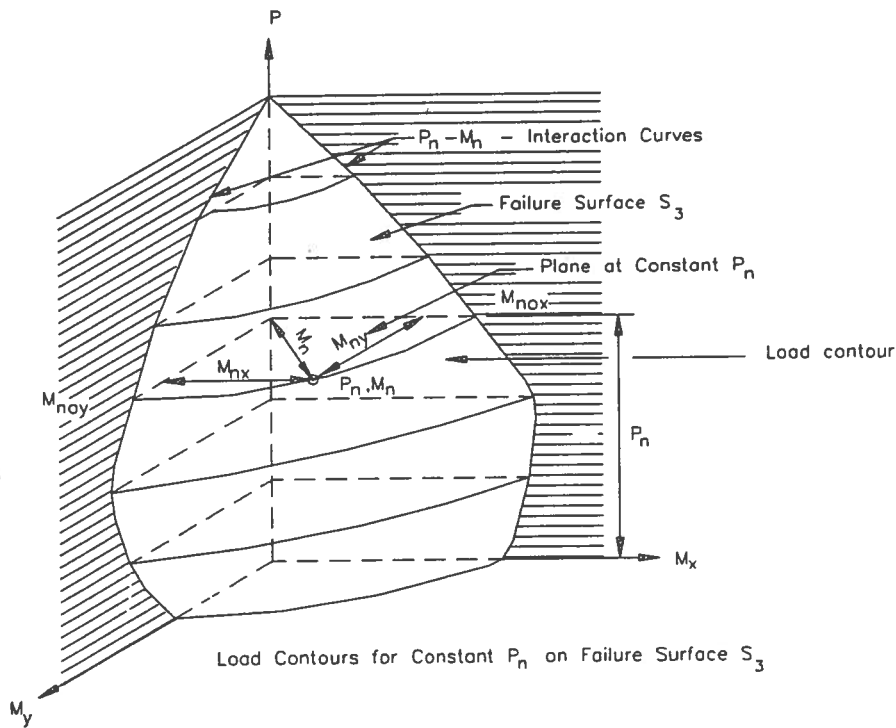
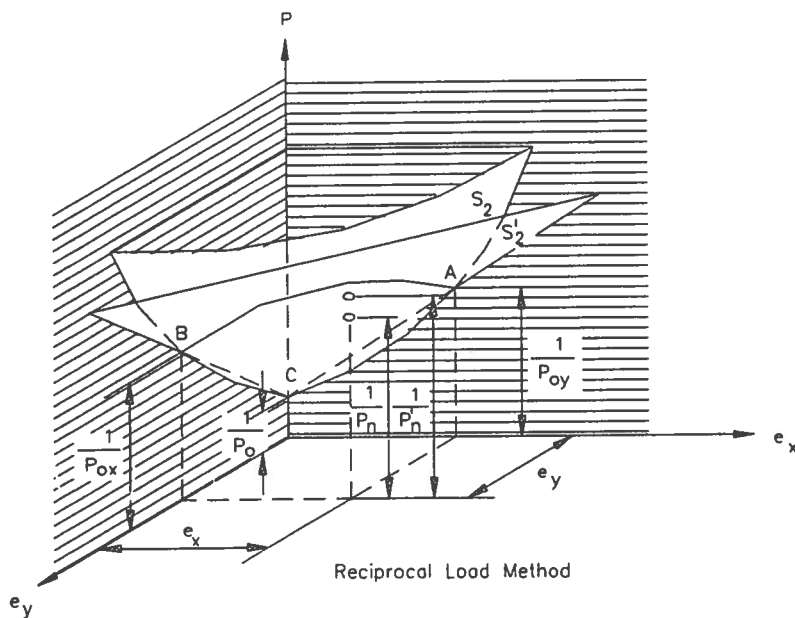


Figure 2. Reciprocal Load Method



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CONCRETE COLUMN DESIGN

Figure 1. Load Contour at Plane of Constant P_u of Failure Surface S_u

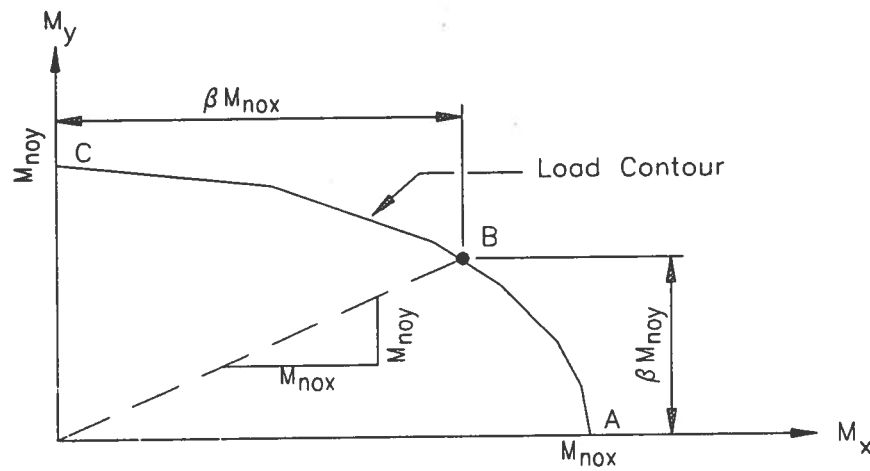
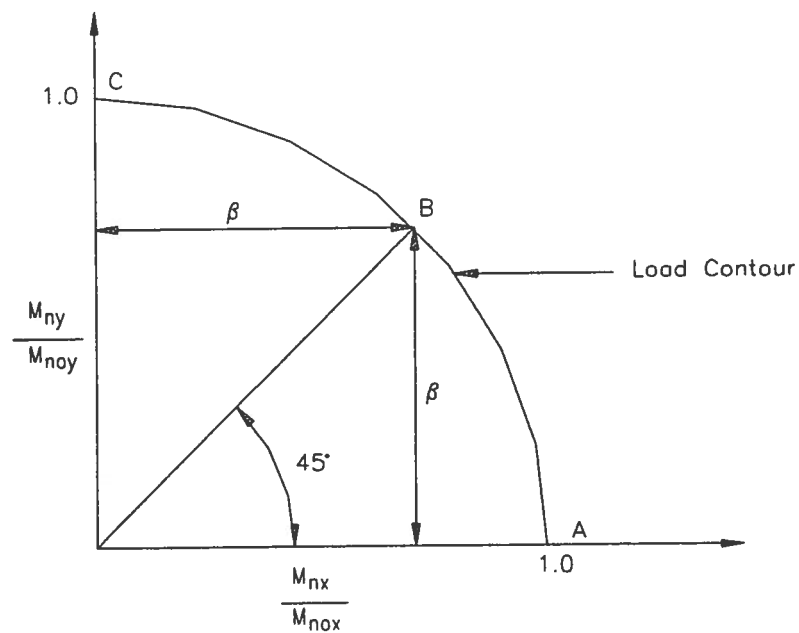


Figure 2. Nondimensional Load Contour at Constant P_u



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Figure 1. Theoretical Contour Line on Interaction Surface

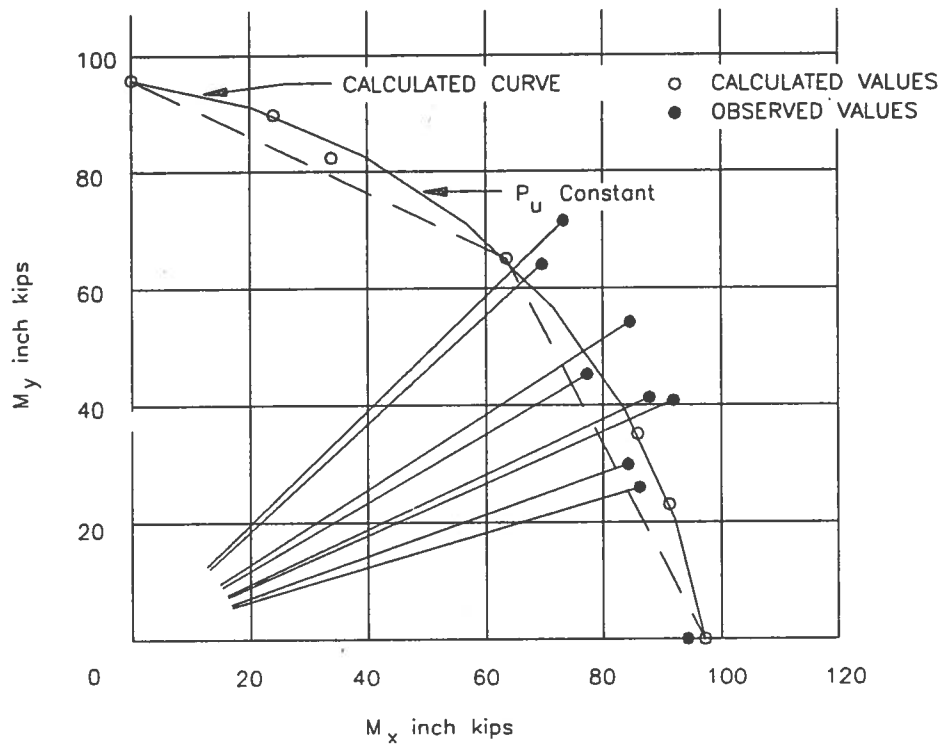
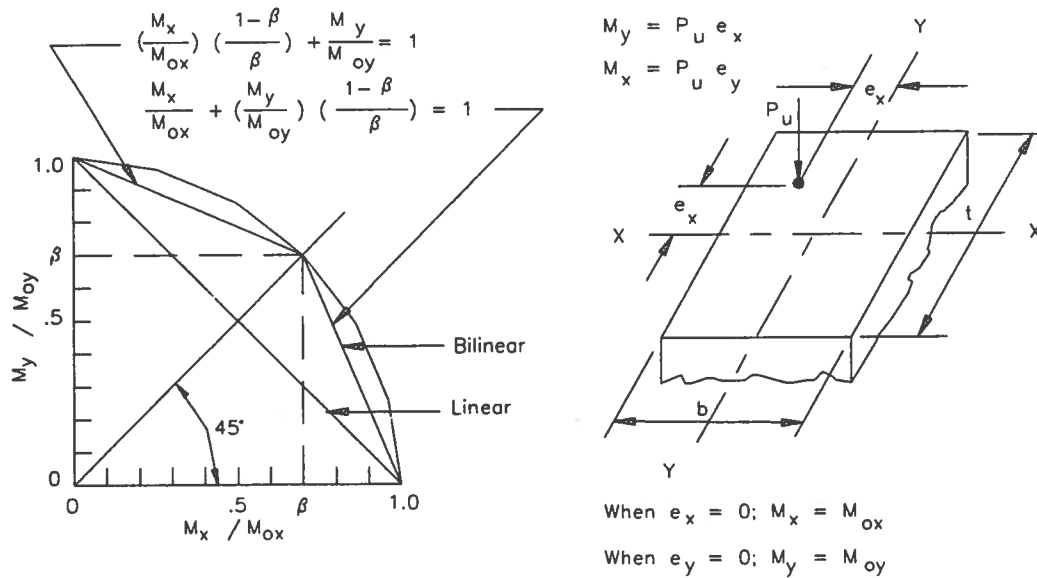
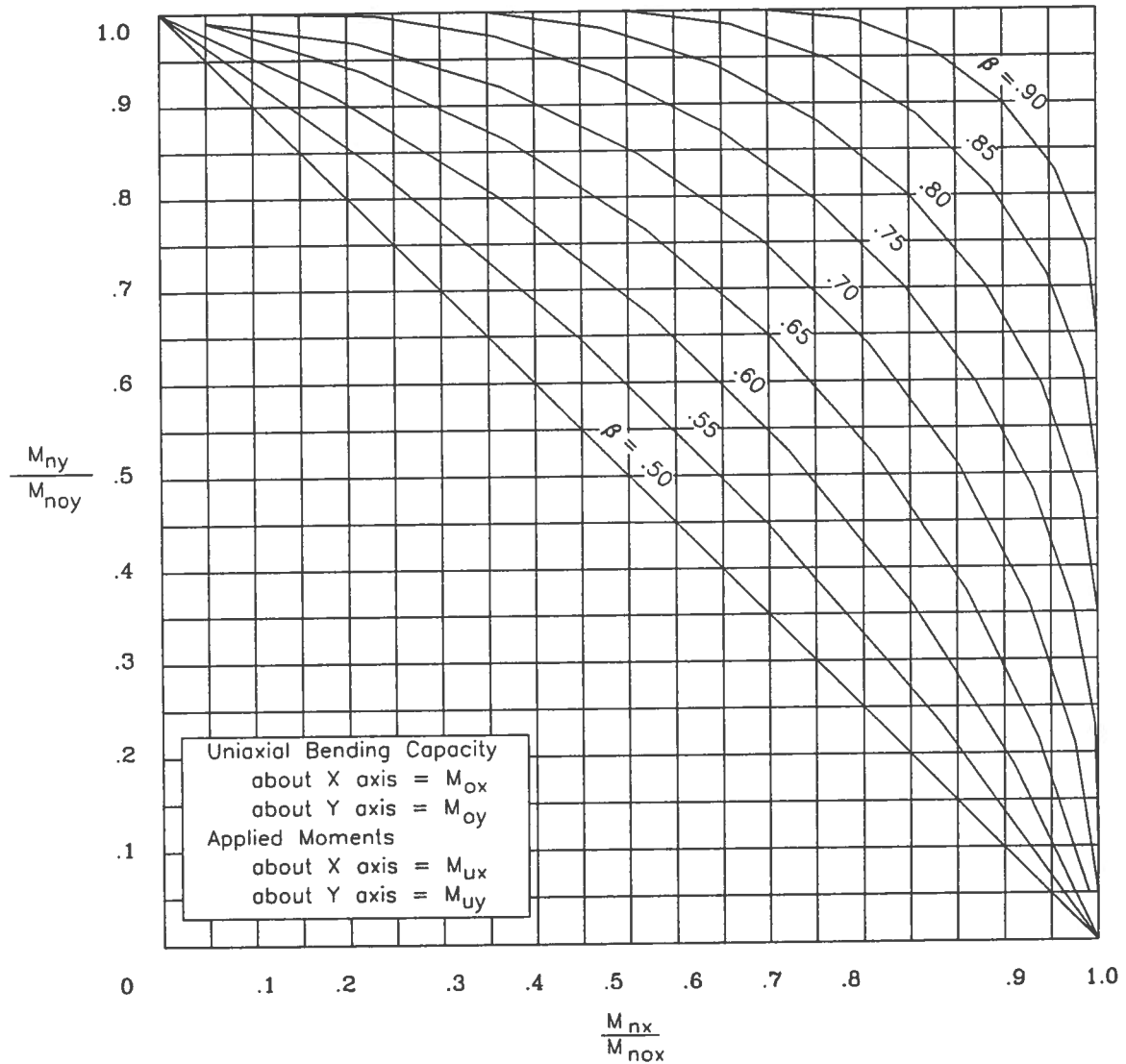


Figure 2. Bilinear Approximation of Load Contour Line



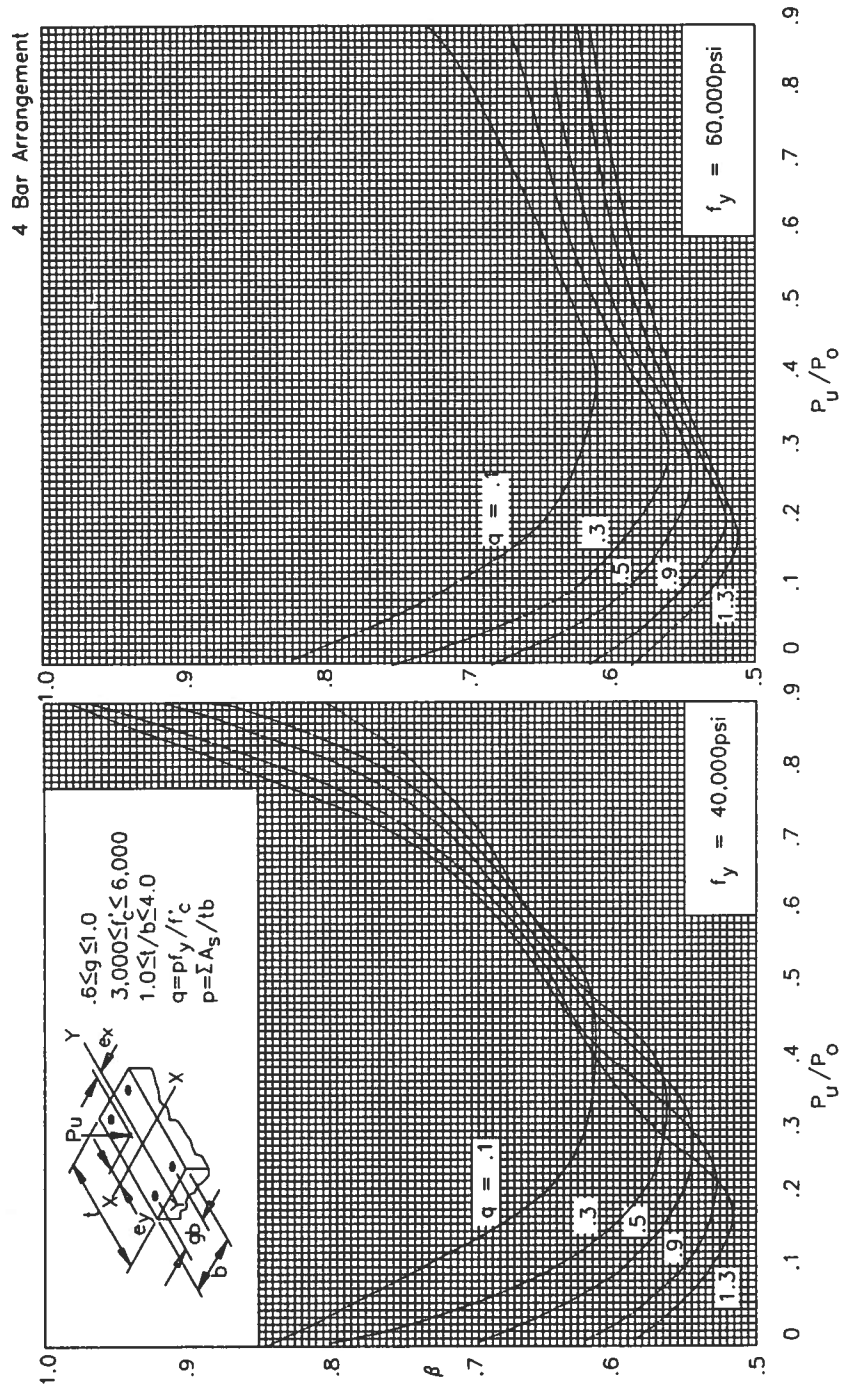
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NONDIMENSIONAL FAILURE CURVES FOR VARIOUS VALUES OF β



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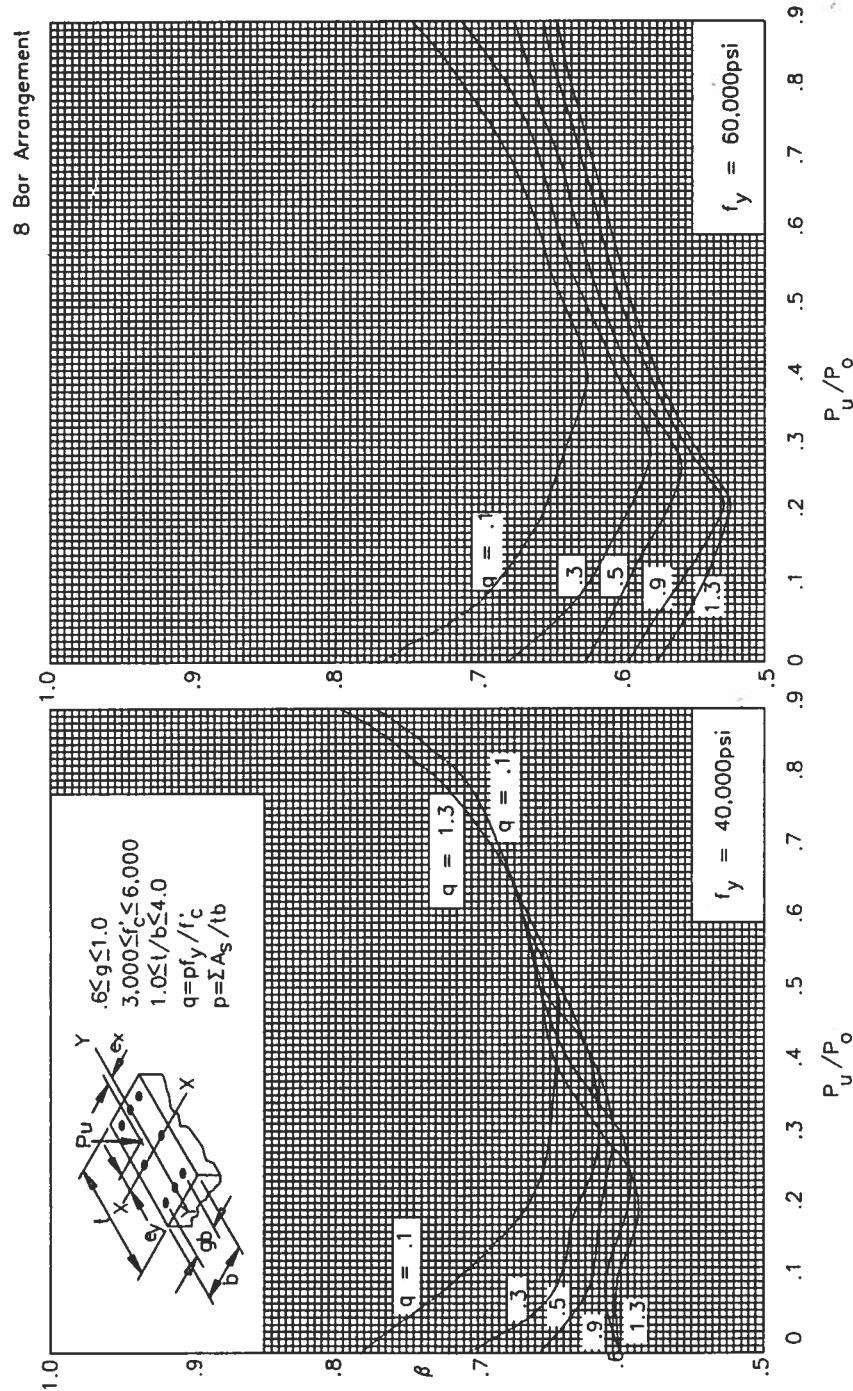
BIAXIAL BENDING DESIGN CONSTANTS



Note: Replace P_u with $\frac{P_u}{\phi}$ and P_o with P_{no} .

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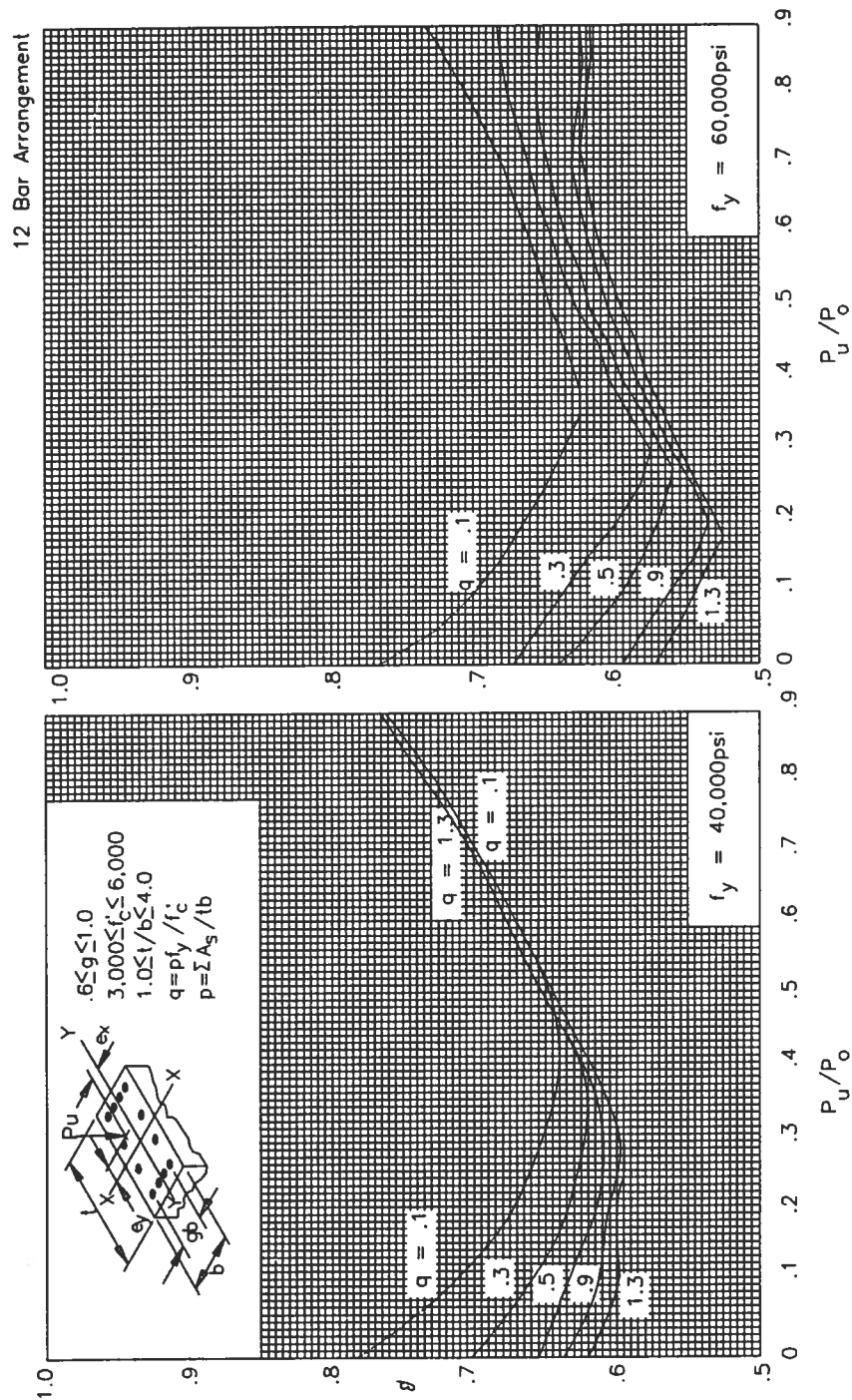
BIAXIAL BENDING DESIGN CONSTANTS



Note: Replace P_u with $\frac{P_u}{\phi}$ and P_o with P_{no} .

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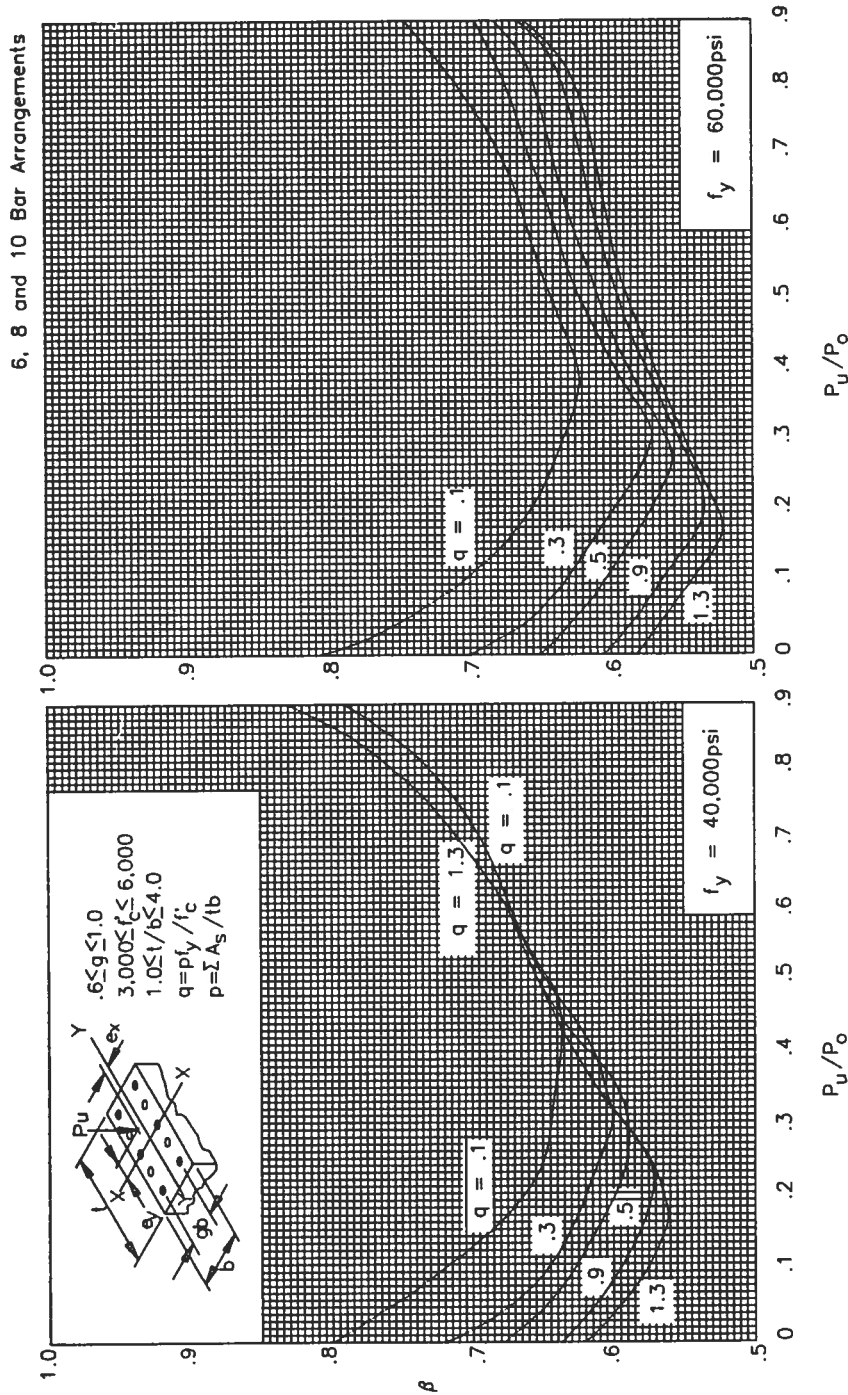
BIAXIAL BENDING DESIGN CONSTANTS



Note: Replace P_u with $\frac{P_u}{\phi}$ and P_o with P_{no} .

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BIAXIAL BENDING DESIGN CONSTANTS



Note: Replace P_u with $\frac{P_u}{\phi}$ and P_o with P_{no} .

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GENERAL NOTES FOR THE SAMPLE DESIGNS

1. The Sample Designs are as follows:
 - Sample Design No. 1 - Trial Section
 - Sample Design No. 2 - Biaxial Bending and Axial Compression
 - 2A Load Contour Method, Spreadsheet Type Calculation
 - 2B Load Reciprocal Method, Spreadsheet Type Calculation
 - 2C Load Contour Method - ACI Handbook
 - 2D Load Reciprocal Method - ACI Handbook
 - Sample Design No. 3 - Biaxial Bending and Axial Tension
 - Sample Design No. 4 - Biaxial Bending Slender Columns
 - Sample Design No. 5 - Uniaxial Bending and Axial Compression of Rectangular Section
 - 5A Spreadsheet Type Calculation
 - 5B ACI - Handbook
 - Sample Design No. 6 - Circular Column
2. The Sample Designs have been chosen, for the most part, using the same loadings so that the various methods and column types might be compared.
3. Concrete strength $f'_c = 4 \text{ ksi}$
Young's Modulus $E_c = 57000 \sqrt{f'_c} = 3600 \text{ ksi}$
4. Yield Strength of reinforcing steel $f_y = 60 \text{ ksi}$
Young's Modulus $E_s = 29000 \text{ ksi}$
5. Sections and equations referenced from ACI 318-89 are quoted as follows:
 - ACI Section . . .
 - ACI Equation . . .
6. Tables, charts, etc. referenced from ACI 340.2r-85 Design Handbook Volume 2 are referenced as follows:
 - ACI-HB . . .
7. Figures and equations referenced from the Narrative section of this Practice are stated simply as:
 - Fig. . . .
 - Eqn. . . .
8. The method of obtaining a trial section demonstrated by Sample Design No. 1, applies to all the Sample Designs modified as follows:
 - Sample Design No. 2
 - No modification

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GENERAL NOTES FOR THE SAMPLE DESIGNS

- Sample Design No. 3

- Modify $A_{ST} = A_{st} + \frac{T_u}{\phi f_y}$

- Sample Design No. 4 (Slenderness)

- The trial section of Sample Design No. 1 will give a preliminary trial section.
 - The preliminary trial section is then refined to produce the trial section for further detailed analysis.

- Sample Design No. 5

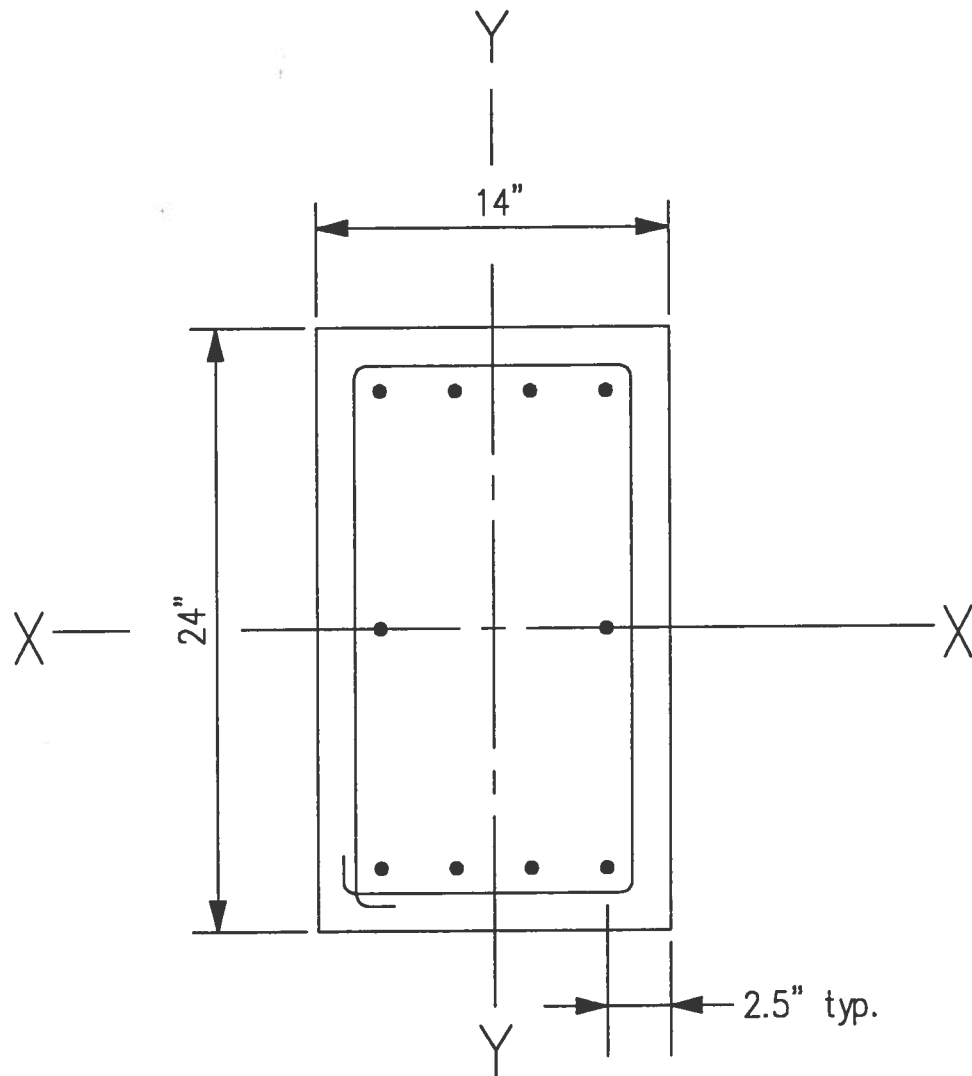
- No modification to procedure

- Sample Design No. 6

- Equivalent circular section is found with same area as square section $h = b$.

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SAMPLE DESIGN 1: BIAXIAL BENDING - TRIAL SECTION



10 #8 BARS

SKETCH 1

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SAMPLE DESIGN 1: BIAXIAL BENDING - TRIAL SECTION

$$M_{ux} = 200^k$$

$$M_{uy} = 85^k$$

$$P_u = 100^k$$

$$f'_c = 4 \text{ ksi}$$

$$f_y = 60 \text{ ksi}$$

Short Column . . . No Slenderness

h / b is not given

$$\therefore \text{Use } h / b = M_{ux} / M_{uy} = 200 / 85 = 2.35 \text{ but not greater than } 2.0$$

$$\therefore \text{Use } h / b = 2.0$$

$$\text{Eqn. (14)} \quad M_{uox} = 1.6 \sqrt{200^{1.6} + (2 \times 85)^{1.6}} = 285.9^k$$

$$\text{Eqn. (15)} \quad h^2 b = (93) \frac{(285.9 \times 12)}{60} = 5317.7$$

$$h / b = 2.0$$

$$h^3 = (5317.7 \times 2); \therefore h = 22.0", \quad b = 11.0"$$

$$\text{Eqn. (16)} \quad A_{st} = 0.03(22.0)(11.0) = 7.26 \text{ in}^2$$

At this point, have the spreadsheet print out $(h, b, A_{st}) = (22.0, 11.0, 7.26)$.

(Note: Decimal places for h and b should be carried, ie. no rounding off)

After above printout, the designer could proceed as follows:

$$A_{st} = 7.26 \text{ in}^2$$

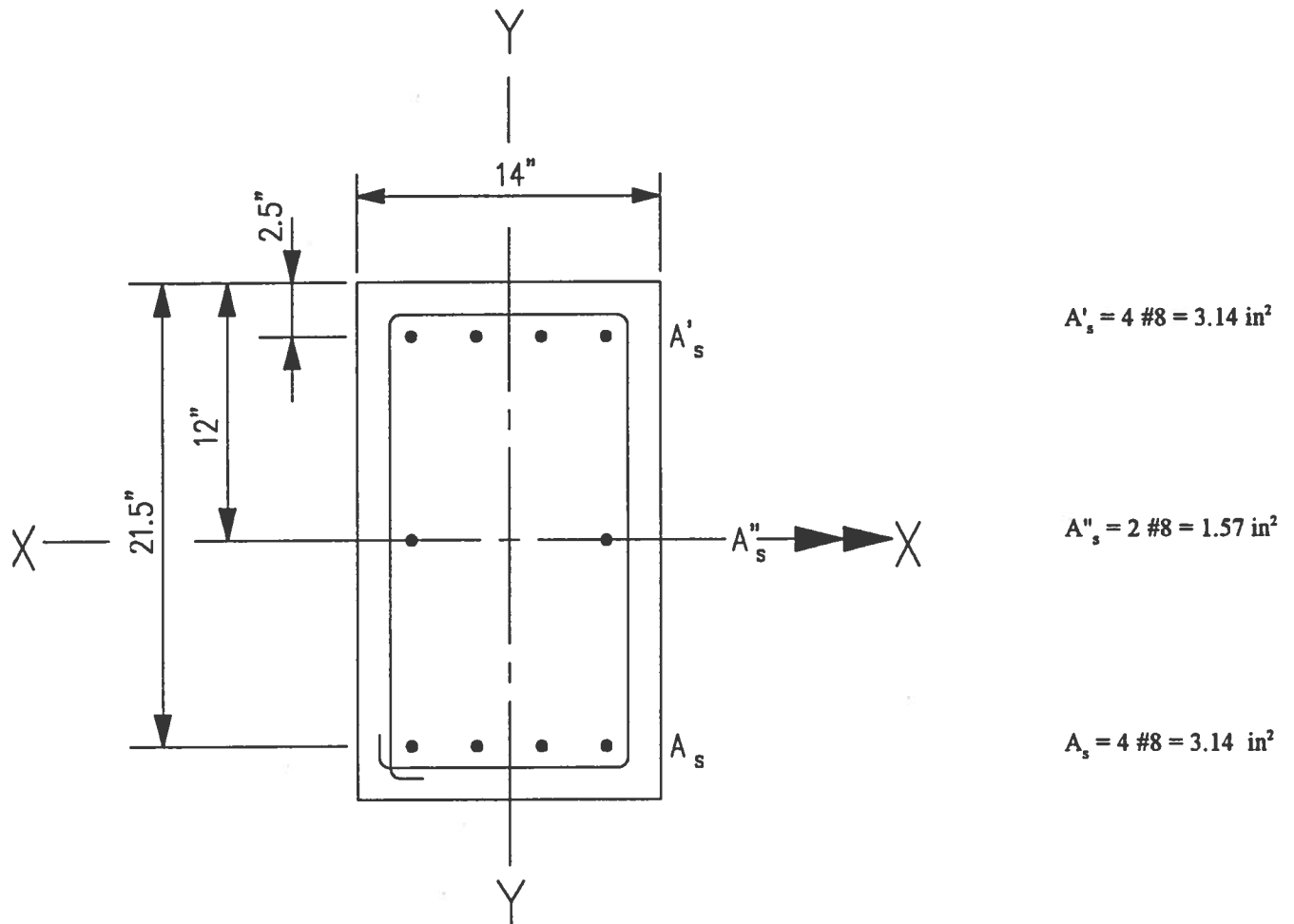
$$\text{Use } 10 \text{ \#8 bars} = 7.85 \text{ in}^2$$

For practical bar spacing, use $b = 14"$ (See ACI-HB, Reinforcement 2, dimension b_2)

Section chosen for further analysis is shown on Sketch 1.

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SAMPLE DESIGN 2: BIAxIAL BENDING AND AXIAL COMPRESSION

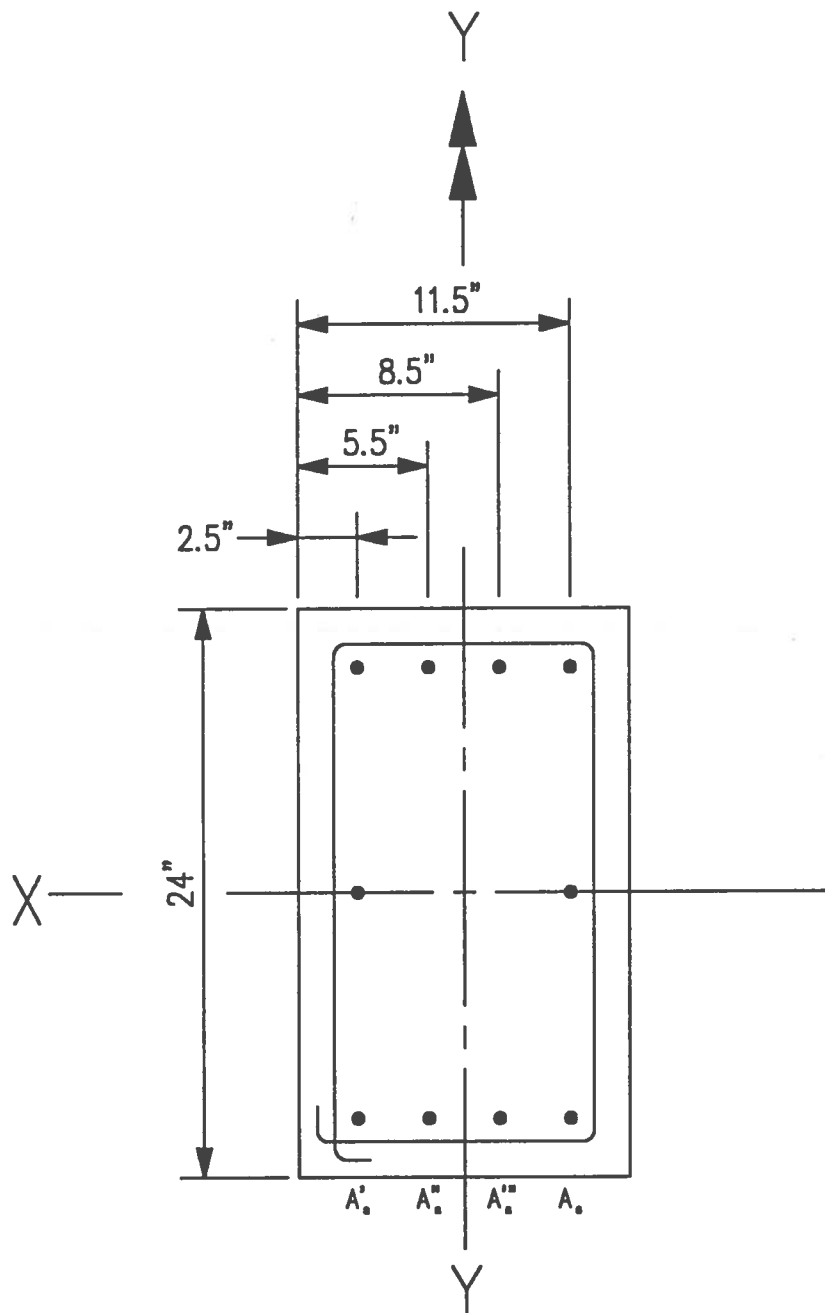


10 #8 BARS

SKETCH 1

FLUOR DANIEL

SAMPLE DESIGN 2: BIAxIAL BENDING AND AxIAL COMPRESSION



$$A'_s = 2.36 \text{ sq in}$$

$$A''_s = 1.57 \text{ sq in}$$

$$A'''_s = 1.57 \text{ sq in}$$

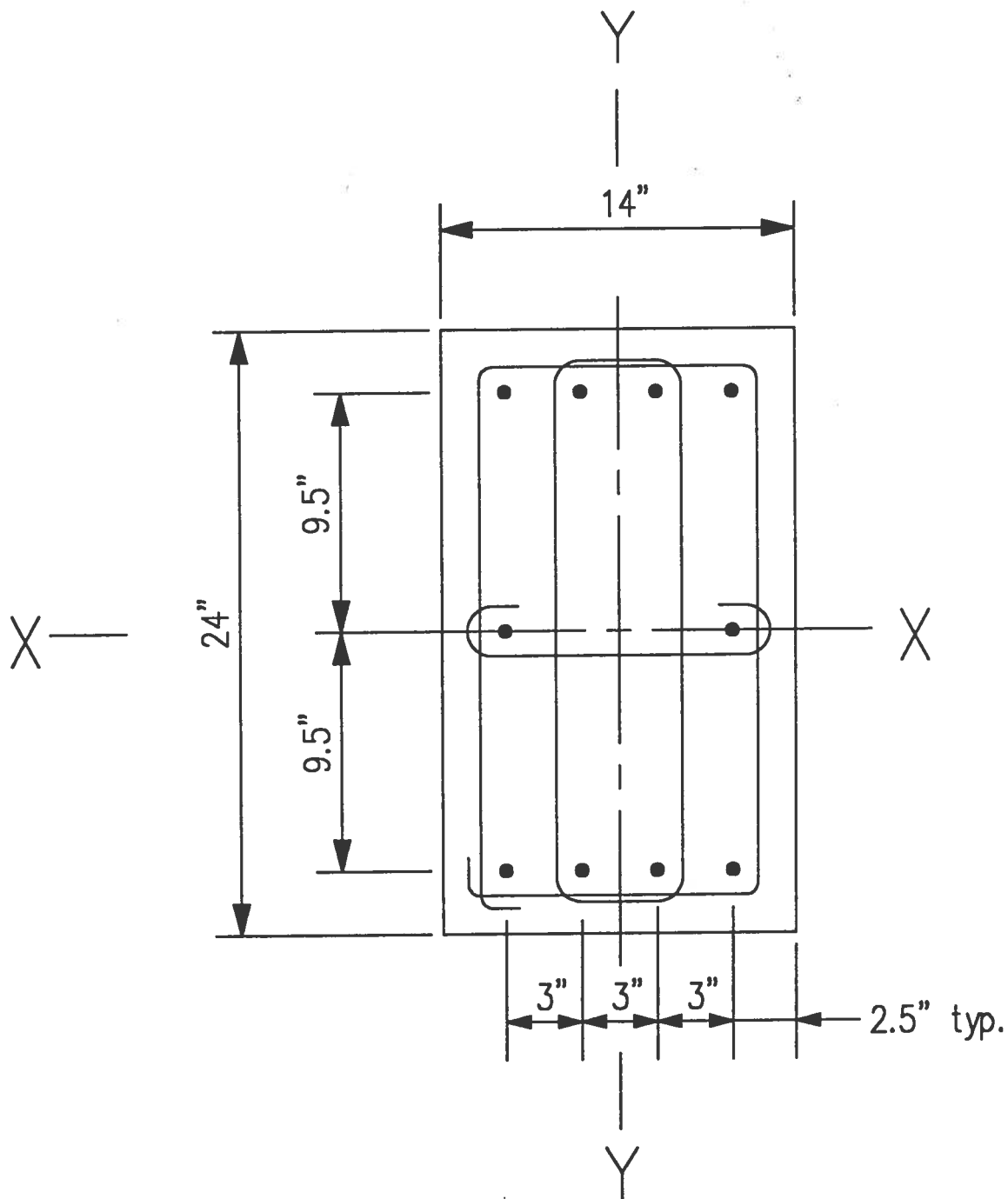
$$A_s = 2.36 \text{ sq in}$$

10 #8 BARS

SKETCH 2

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION



10 #8 BARS WITH #3 TIES @ 14" O/C (SETS OF 3)

SKETCH 3

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SAMPLE DESIGN 2: BIAxIAL BENDING AND AXIAL COMPRESSION

Use the applied loads of example 1, and the trial section shown on Page 1 of Sample Design 1.

2A - LOAD CONTOUR METHOD - SPREADSHEET TYPE CALCULATION

Find M_{nox} at P_u / ϕ :

For tied columns, $\phi = 0.7$

ACI Section 9.3.2.2

$$P_u / \phi = 100 / 0.7 = 142.9^k$$

Refer to figs. 1 and 2, and apply eqns (1) through (8)

At balanced condition

$$\text{Eqn. (8)} \quad c_b = \frac{0.003}{0.003 + e_y} d$$

$$e_y = f_y / E_s = 60 / 29000 = 0.002069$$

$$d = 24 - 2.5 = 21.5 \text{ for bending about the X-axis}$$

$$\therefore c_b = \frac{0.003}{0.003 + 0.002069} (21.5) = 12.72"$$

$$\text{Eqn. (1)} \quad e'_s = \left(\frac{12.72 - 2.5}{12.72} \right) (0.003) = 0.0024104$$

$$e''_s = \left(\frac{12.72 - 12}{12.72} \right) (0.003) = 0.0001698$$

$$e_s = e_y = 0.00207$$

$$\text{Eqn. (2)} \quad f'_s = 29000(0.0024104) = 69.9\text{ksi} > f_y = 60\text{ksi, ie. compression steel yields}$$

$$f''_s = 29000(0.0001698) = 4.92\text{ksi (compression)}$$

$$f_s = f_y = 60 \text{ ksi}$$

$$\text{Eqn. (3)} \quad C' = 60 \times 3.14 = 188.4^k$$

$$C'' = 1.57 \times 4.92 = 7.72^k$$

$$T = 60 \times 3.14 = 188.4^k$$

$$\text{Eqn. (4a)} \quad C = 0.85 f'_c b a = 0.85 f'_c b (\beta_1 c)$$

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

Use $\beta_1 = 0.85$ for $f'_c = 4 \text{ ksi}$ & $f_y = 60 \text{ ksi}$

$$\therefore C = 0.85(4)(14)(0.85 \times 12.72) = 514.7^k$$

Eqn. (6) $P + \Sigma T = \Sigma C$

$$P = 188.4 + 514.7 + 7.72 - 188.4 = 522.4^k$$

ie. At balanced condition, $P = 522.4 \text{ k} > P_u / \phi = 142.9^k$

$$\therefore c < c_b = 12.72$$

Try $c = 6''$ and repeat calculation.

$$C' = 159.4^k$$

$$T'' = 94.2^k \text{ ie. } A_s \text{ is now in Tension}$$

$$T = 188.4^k$$

$$C = 242.8^k$$

$$\therefore P = 119.6^k < P_u / \phi = 142.9^k$$

$$\therefore c > 6''$$

Try $c = 6.4$ and repeat calculation.

$$C' = 166.5^k$$

$$T'' = 94.2^k$$

$$T = 188.4^k$$

$$C = 258.9^k$$

$$\therefore P = 142.8^k \cong P_u / \phi = 142.9^k$$

Close enough $\therefore$ Use $P_n = 142.8^k$

Take moments about tension steel A_s

Eqn. (5) $l'_s = 21.5 - 2.5 = 19''$

$$l''_s = 21.5 - 12 = 9.5''$$

$$l_s = 21.5 - \frac{0.85(6.4)}{2} = 18.78''$$

Eqn. (7) $142.8(e') = (166.5)(19) + (258.9)(18.78) - (94.2)(9.5)$

$$\therefore e' = 49.97''$$

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

$$\therefore e_{oy} = e' - d + \frac{h}{2} = 49.97 - 21.5 + \frac{24}{2} = 40.47"$$

$$\therefore M_{nox} = P_n e_{oy} = (142.8^k)(40.47"/12) = 481.6^k$$

Find M_{noy} at $P_u / \phi = 142.9^k$

Again, refer to figures 1 and 2 , and apply Eqns. (1) through (8). Repeating similar calculation as above:

At balanced condition,

$$\text{Eqn. (8)} \quad c_b = \frac{0.003}{0.003 + 0.002069}(11.5) = 6.81"$$

$$\begin{aligned} \therefore C' &= 129.9^k \\ C'' &= 26.5^k \\ T'' &= 34.0^k \\ T &= 141.6^k \end{aligned}$$

$$\text{Eqn. (4b)} \quad C = 0.85 f'_c h a = 0.85 f'_c h (0.85 \times 6.81) = 0.85(4)(24)(0.85 \times 6.81) = 472.3^k$$

$$\text{Eqn. (6)} \quad P = 452.8 > P_u / \phi = 142.9^k$$

$$\therefore c < c_b = 6.81"$$

Try $c = 4.0"$ and repeat calculation

$$\begin{aligned} C' &= 77.0^k \\ T'' &= 51.2^k \text{ (changed from compression to tension)} \\ T''' &= 94.2^k \\ C &= 277.4^k \end{aligned}$$

$$\therefore P = 67.4^k < P_u / \phi = 142.9^k$$

$$\therefore c > 4"$$

Try $c = 4.5"$ and repeat calculation

$$\begin{aligned} C' &= 91.3^k \\ T'' &= 30.4^k \text{ (changed from compression to tension)} \\ T''' &= 94.2^k \\ T &= 141.6^k \\ C &= 312.1^k \end{aligned}$$

$$\therefore P = 137.2^k \text{ just } < P_u / \phi = 142.9^k$$

$$\therefore c \text{ just } > 4.5"$$

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

Try $c = 4.55"$ and repeat calculation

$$\begin{aligned} C' &= 92.5^k \\ T'' &= 28.5^k \text{ (changed from compression to tension)} \\ T''' &= 94.2^k \\ T &= 141.6^k \\ C &= 315.6^k \end{aligned}$$

$$\therefore P = 143.8^k \cong P_u / \phi = 142.9 = P_n$$

Take moments about tension steel A_s

$$\text{Eqn. (5)} \quad l'_a = 11.5 - 2.5 = 9"$$

$$l''_a = 11.5 - 5.5 = 6"$$

$$l_a = 11.5 - \frac{0.85(4.55)}{2} = 9.57"$$

$$\text{Eqn. (7)} \quad 143.8 (e') = (92.5)(9) + (315.6)(9.57) - (28.5)(6) - (94.2)(3)$$

$$\therefore e' = 23.64"$$

$$\therefore e_{ox} = e' - d + \frac{b}{2} = 23.64 - 11.5 + \frac{14}{2} = 19.14"$$

$$\therefore M_{noy} = P_n e_{ox} = (143.8^k)(19.14"/12) = 229.4^k$$

$$\therefore M_{nox} = 481.6^k$$

$$M_{noy} = 229.4^k \text{ at } P_n = P_u / \phi = 142.9^k$$

Find β

From ACI Section 10.3.5.2,

$$\begin{aligned} P_{no} &= 0.80 [0.85 f'_c (A_g - A_{st}) + f_y A_{st}] \\ &= 0.80 [0.85(4)((14 \times 24) - 7.85) + (60)(7.85)] = 1269^k \end{aligned}$$

$$\therefore P_n / P_{no} = 142.9 / 1269 = 0.113$$

$$q = \left(\frac{A_s}{h b} \right) \left(\frac{f'_c}{f_y} \right) = \left(\frac{7.85}{14 \times 24} \right) \left(\frac{60}{4} \right) = 0.35$$

From fig. 11 for 10 bar arrangement, $\beta = 0.62$

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

Eqn. (17) $\therefore N = \frac{\log 0.50}{\log 0.62} = 1.45$

Using Eqn. (18), $\left[\frac{200/0.7}{481.2} \right]^{1.45} + \left[\frac{85/0.7}{229.4} \right]^{1.45} = 0.867 < 1.0$

$\therefore$ TRIAL SECTION IS ADEQUATE AS SHOWN ON SKETCH 1 OF SAMPLE DESIGN 2

Lateral Reinforcement

Lateral reinforcement in this Practice is in accordance with ACI section 7.10.5, and will be designed only for this example.

For special requirements, eg. when shear and/or torsion occur, seismic detailing, etc. see practices listed in Attachment No. 2.

ACI Section 7.10.5.1 Use #3 ties

ACI Section 7.10.5.2 Vertical spacing $\leq 48 \times 3/8 = 18"$
or $\leq b = 14"$
or $\leq 16 \times 1 = 16"$

$\therefore$ Use 14" vertical spacing.

ACI Section 7.10.5.3 See Sketch 3 of Sample Design 2 for one arrangement that satisfies this section of ACI.

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SAMPLE DESIGN 2: BIAxIAL BENDING AND AXIAL COMPRESSION

2B - LOAD RECIPROCAL METHOD - SPREADSHEET TYPE CALCULATION

Check if $P_u \geq 0.1 f'_c h b$

$$P_u / \phi = 100 / 0.7 = 142.9^k$$

$$0.1 f'_c h b = 0.1(4)(24)(14) = 134.4^k$$

$$\therefore P_u / \phi > 0.1 f'_c h b$$

$\therefore$ Feasible to use Load Reciprocal Method

Find P_{max} at $e_y = M_{ux} / P_u$

$$e_y = M_{ux} / P_u = (200 \times 12) / 100 = 24"$$

At balanced condition (see ex. 2A)

$$C' = 188.4^k$$

$$C'' = 7.72^k$$

$$T = 188.4^k$$

$$C = 514.7^k$$

As per Ex. 2A

$$\therefore P = 522.4^k$$

Take moments about tension steel A_s

$$\text{Eqn. (5)} \quad l'_s = 21.5 - 2.5 = 19"$$

$$l''_s = 21.5 - 12 = 9.5"$$

$$l_s = 21.5 - \frac{0.85(12.72)}{2} = 16.09"$$

$$\text{Eqn. (7)} \quad 522.4 (e') = (188.4)(19) + (514.7)(16.09) + (7.72)(9.5)$$

$$\therefore e' = 22.85"$$

$$\therefore e = 22.85 - 21.5 + \frac{24}{2} = 13.35"$$

$$\therefore e_y = 24" > e = 13.35"$$

$$\therefore c < c_b = 12.72"$$

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

Try $c = 8"$ and repeat calculations similar to ex. 2A

Using Eqns (1) to (7):

$$C' = 187.8^k$$

$$T'' = 68.3^k$$

$$T = 188.4^k$$

$$C = 323.7^k$$

$$\therefore P = 254.8^k$$

$$I'_a = 19"$$

$$I''_a = 9.5"$$

$$I_a = 21.5 - \frac{0.85(8)}{2} = 18.1"$$

$$254.8(e') = (187.8)(19) + (323.7)(18.1) - (68.3)(9.5)$$

$$\therefore e' = 34.45"$$

$$\therefore e = 34.45 - 21.5 + \frac{24}{2} = 25.0"$$

$$\therefore e_y = 24" \text{ just } < 25"$$

$$\therefore c > 8"$$

Repeating calculation with $c = 8.1"$ gives:

$$P = 262.0^k$$

$$e = 24.4" \approx e_y = 24" \text{ close enough}$$

$$\text{ie. } P = P_{\text{nox}} = 262.0^k \text{ at the given eccentricity } e_x = 24"$$

Find P_{noy} at $e_x = M_{\text{uy}} / P_u$

$$e_x = M_{\text{uy}} / P_u = (85 \times 12) / 100 = 10.2"$$

By repeating the iterative procedure as before, we find that

$$C = 5.6"$$

$$C' = 113.7^k$$

$$C'' = 2.4^k$$

$$T'' = 70.7^k$$

$$T = 141.6^k$$

As per Ex. 2A

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

$$C = 388.4^k$$

$$\therefore P = 292.2^k$$

$$e' = 14.95"$$

and $e = 10.45" = e_x = 10.2"$

$$\therefore P = P_{\text{noy}} = 292.2 \text{ @ } e_x = 10.2"$$

Thus: $P_{\text{nox}} = 262.0^k$

$$P_{\text{noy}} = 292.2^k$$

$$P_{\text{no}} = 1269^k \dots \text{see ex. 2A}$$

From Eqn. (9), $1/P_n = 1/262.0 + 1/292.2 - 1/1269$

$$\therefore P_n = 155.0^k$$

$$\phi P_n = 0.7(155^k) = 108.5 \text{ k} > P_u = 100^k$$

$\therefore$ By Reciprocal Load Method, Section is adequate.

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

2C - LOAD CONTOUR METHOD, USING ACI HANDBOOK

Using the trial section of SKETCH-1 of Sample Design 1, find β from Parme's charts.

$$\therefore \beta = 0.62 \text{ as found before}$$

$$\therefore N = \frac{\log 0.50}{\log 0.62} = 1.45$$

$$\text{Using: } M_{nx} = M_{ux} / \phi = 200 / 0.7 = 285.7 \text{ K}$$

$$M_{ny} = M_{uy} / \phi = 85 / 0.7 = 121.4 \text{ K}$$

$$\text{Eqn. (11b)} \quad M_{nox} = {}^{1.45} \sqrt{(285.7)^{1.45} + \left(\frac{24}{14} \times 121.4\right)^{1.45}} = 400.4 \text{ K}$$

$$\text{For bending about X-axis, } \gamma = \frac{24 - 2(2.5)}{24} = 0.79$$

Use ACI - HB Columns 7.4.3

$$\frac{\phi P_n}{A_g} = \frac{P_u}{A_g} = \frac{100}{24(14)} = 0.298$$

$$\frac{\phi M_n}{A_g h} = \frac{\phi M_{nox}}{A_g h} = \frac{0.7(400.4 \times 12)}{24(14)(24)} = 0.417$$

Entering the chart: $\rho_g = 0.022$

$$\therefore A_{st} = 0.022 \times 14 \times 24 = 7.4 \text{ in}^2$$

$$\therefore A_{st} = 10\text{-}\#8 \text{ bars} = 7.85 \text{ in}^2$$

$\therefore$ Section okay.

As a second alternative, we could proceed:

$$\frac{P_u}{A_s} = 0.298 \text{ as before}$$

$$\text{Reinforcement ratio provided} = \rho_s = \frac{A_s}{A_g} = \frac{7.85}{(24 \times 14)} = 0.023$$

$$\therefore \frac{\phi M_n}{A_s h} = 0.45$$

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SAMPLE DESIGN 2: BIAxIAL BENDING AND AXIAL COMPRESSION

$$\therefore M_n = (0.45) \frac{(24 \times 14)(24)}{0.7} = 5184 \text{ K}'' = 432 \text{ K}'$$

$$\text{ie. } M_{\text{nox}} \text{ provided} = 432 \text{ K}' > 400.4 \text{ K}' \text{ required}$$

And again, as a third alternative, proceed as follows:

$$\text{Find actual } M_{\text{nox}} \text{ of section: } M_{\text{nox}} = 432 \text{ K}' \text{ as found before}$$

$$\text{Find actual } M_{\text{noy}} \text{ of section: } \frac{P_u}{A_g} = 0.298 \text{ as before}$$

$$\rho_g = 0.023 \text{ as before}$$

$$\text{For bending about Y-axis, } \gamma = \frac{14 - 2(2.5)}{14} = 0.64$$

Use ACI-HB Columns 7.4.2 where $\gamma = 0.60$

Note: We may use both Columns 7.4.2 where ($\gamma = 0.60$) and Columns 7.4.3 ($\gamma = 0.75$) and interpolate to get results for $\gamma = 0.64$; in this example we will use Columns 7.4.2 ($\gamma = 0.60$) which will be slightly conservative.

$$\therefore \frac{\phi M_n}{A_g b} = 0.43$$

$$\therefore M_n = (0.43) \frac{(24 \times 14)(14)}{0.7} = 2890 \text{ K}'' = 240.8 \text{ K}'$$

$$\text{ie. } M_{\text{noy}} = 240.8 \text{ K}'$$

$$\text{Thus: } M_{\text{nox}} = 432 \text{ K}'$$

$$M_{\text{noy}} = 240.8 \text{ K}'$$

$$N = 1.45$$

$$\text{Eqn. (18)} \quad \left(\frac{200/0.7}{432} \right)^{1.45} + \left(\frac{85/0.7}{240.8} \right)^{1.45} = 0.920$$

Note: That the second alternative procedure gave:

$$\text{Required } M_{\text{nox}} / \text{Actual } M_{\text{nox}} = 400.4 / 432 = 0.927$$

Compare with unity check above.

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

2D - LOAD RECIPROCAL METHOD, USING ACI HANDBOOK

Use trial section of SKETCH 1, Sample Design 1.

In example 2B we see that $P_u / \phi > 0.1 f'_c h b$

$\therefore$ Load Reciprocal Method maybe used.

P_{no} : This was already found in example 2A to be 1269^K

$$P_{nox}: \frac{e_y}{h} = \frac{M_{ux}}{P_u h} = \frac{200 \times 12}{(100)(24)} = 1.0$$

$$A_{st} = 10 \#8 = 7.85 \text{ in}^2$$

$$\therefore \rho_g = \frac{7.85}{(24 \times 14)} = 0.023$$

$$\text{For bending about x-axis, } \gamma = \frac{24 - 2(2.5)}{24} = 0.79$$

... Use ACI-HB Columns 7.4.3

$$\frac{e_y}{h} = 1.0; \rho_g = 0.023$$

$$\therefore \frac{\phi P_n}{A_g} = 0.45$$

$$\therefore P_n = 0.45 \left(\frac{24 \times 14}{0.7} \right) = 216^K$$

$$\text{ie. } P_{nox} = 216^K$$

$$P_{noy}: \frac{e_x}{b} = \frac{M_{uy}}{P_u b} = \frac{85 \times 12}{(100)(14)} = 0.73$$

$$\rho_g = 0.023 \text{ as above}$$

$$\text{For bending about the Y-axis, } \gamma = \frac{14 - 2(2.5)}{14} = 0.64$$

Use ACI-HB Columns 7.4.2

where $\gamma = 0.60$. This will be slightly conservative.

$$\therefore \frac{\phi P_n}{A_g} = 0.65$$

$$\therefore P_n = (0.65) \frac{(24 \times 14)}{0.7} = 312^K$$

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SAMPLE DESIGN 2: BIAXIAL BENDING AND AXIAL COMPRESSION

Eqn. (9) $1/P_n = 1/216 + 1/312 - 1/1269$

$$\therefore P_n = 142^k, \quad \phi P_n = 99.3^k \quad \text{just} < \quad P_u = 100^k$$

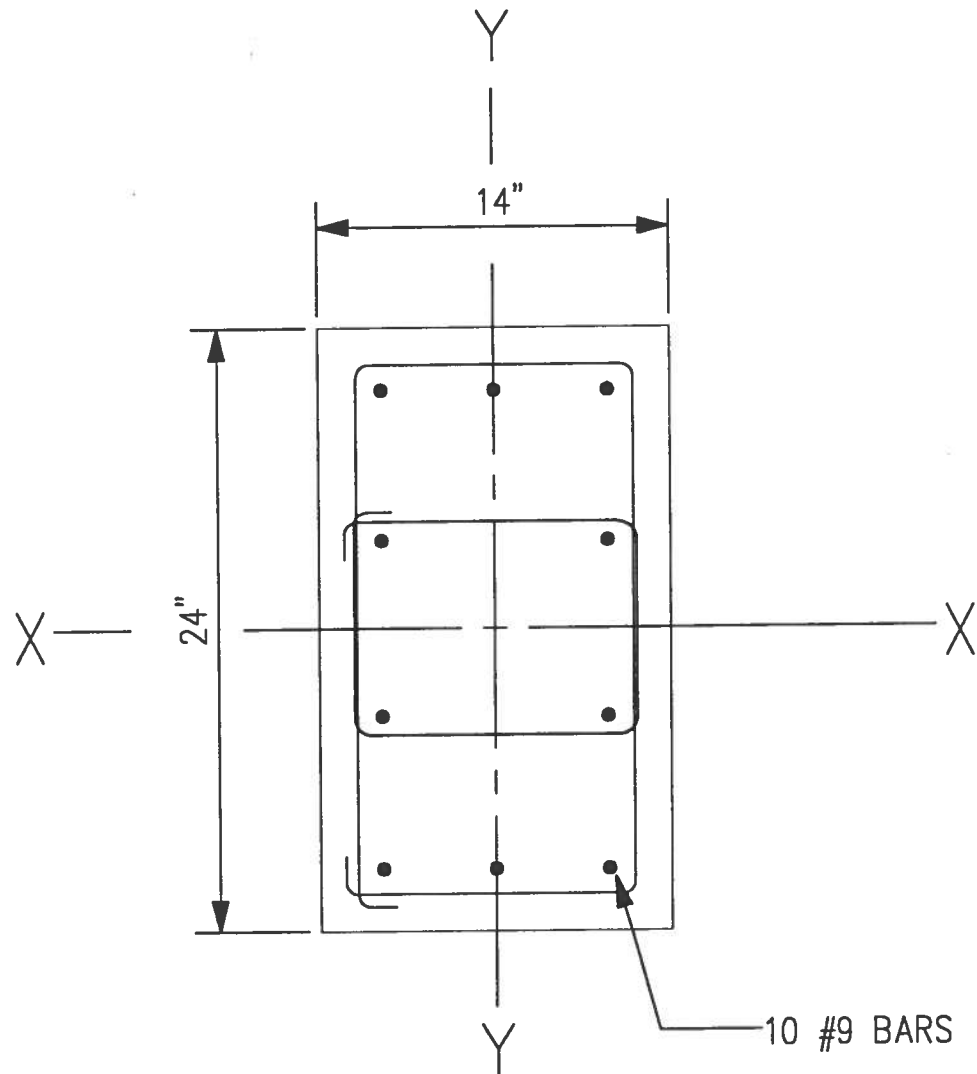
Section is okay since chart used with $\gamma = 0.60$ is slightly conservative. Also, this shows the conservative nature of the Load Reciprocal Method, as discussed in the ACI 318 Code Commentary.

SUMMARY: Comparison of results of Sample Designs 2A to 2D:

2A:	Unity check = 0.906	(Load Contour - Spreadsheet Calculations)
2B:	$\frac{P_u}{\phi P_n} = \frac{100}{108.5} = 0.922$	(Load Reciprocal - Spreadsheet Calculations)
2C:	Unity check = 0.944	(Load Contour - ACI Handbook)
2D:	$\frac{P_u}{\phi P_n} = \frac{100}{99.3} = 1.007$	(Load Reciprocal - ACI Hamdbook)

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SAMPLE DESIGN 3: BIAXIAL BENDING AND AXIAL TENSION



TIES - #3 @ 14" O/C (SETS OF 2)

SKETCH 1

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SAMPLE DESIGN 3: BIAXIAL BENDING AND AXIAL TENSION

$$M_{ux} = 200^k$$

$$M_{uy} = 85^k$$

$$T_u = 150^k$$

$$f'_c = 4 \text{ ksi}$$

$$f_y = 60 \text{ ksi}$$

Determine an initial trial section as per example 1 . . .

Check this section for pure biaxial bending, i.e. $P = 0$ as follows:

Find M_{nox} at $P_n = 0$

This can be accomplished using spreadsheet type calculation, or by using ACI-HB charts. Spreadsheet calculation will be used in this example.

$$\text{Eqn. (6)} \quad P + \Sigma T = \Sigma C, \quad P = 0, \quad \therefore \Sigma T = \Sigma C$$

At balanced condition

$$c_b = \frac{0.003}{0.003 + 0.002069} (21.5) = 12.72" \text{ as before.}$$

$$\therefore C' = 188.4^k$$

$$C'' = 7.73^k$$

$$C = 514.7^k$$

$$T = 188.4^k$$

$$\Sigma T = 188.4^k$$

$$\Sigma C = 710.8^k$$

$$\therefore \Sigma C > \Sigma T$$

$$\therefore \text{required } c < c_b = 12.72"$$

After a couple of iterations, we find that at $c = 4.2"$,

$$C' = 110.6^k$$

$$T'' = 94.2^k$$

$$C = 169.9^k$$

$$T = 188.4^k$$

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SAMPLE DESIGN 3: BIAxIAL BENDING AND AXIAL TENSION

$$\Sigma T = 282.6^k$$

$$\Sigma C = 280.5^k$$

$$\therefore \Sigma T \cong \Sigma C \text{ close enough at } c = 4.2''$$

Taking moments about tension steel A_s

$$l'_a = 21.5 - 2.5 = 19''$$

$$l''_a = 21.5 - 12 = 9.5''$$

$$l_a = 21.5 - \frac{0.85(4.2)}{2} = 19.72''$$

$$\therefore M_{\text{tox}} = C' l'_a + C l_a - T'' l''_a = (110.6)(19) + (169.9)(19.72) - (94.2)(9.5) = 4557^k = 379.7^k$$

Find M_{toy} at $P_n = 0$

Repeating above calculation, bending about the Y-axis, we get (after the second iteration):

$$c = 3.56''$$

$$C' = 61.1^k$$

$$T'' = 74.4^k$$

$$T''' = 94.2^k$$

$$T = 141.6^k$$

$$C = 246.9^k$$

$$\Sigma C = 308.0^k$$

$$\Sigma T = 310.2^k$$

$$\therefore \Sigma T \cong \Sigma C \text{ at } c = 3.56''$$

Taking moments about tension steel A_s ,

$$l'_a = 11.5 - 2.5 = 9''$$

$$l''_a = 11.5 - 5.5 = 6''$$

$$l'''_a = 11.5 - 8.5 = 3''$$

$$l_a = 11.5 - \frac{0.85(3.56)}{2} = 9.99''$$

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SAMPLE DESIGN 3: BIAXIAL BENDING AND AXIAL TENSION

$$\begin{aligned}\therefore M_{\text{noy}} &= C' I'_a + C I_a - T'' I''_a - T''' I'''_a \\ &= (61.1)(9) + (246.9)(9.99) - (74.4)(6) - (94.2)(3) = 2287.4 \text{ k}'' = 190.6 \text{ k}''\end{aligned}$$

$\therefore$ For pure bending, at $P_u = 0$,

$$M_{\text{nox}} = 379.7 \text{ k}''$$

$$M_{\text{noy}} = 190.6 \text{ k}''$$

Find β

$$\frac{(P_u/\phi)}{P_{\text{no}}} = 0, \text{ since } P_u = 0$$

$$q = \left(\frac{A_s}{h b}\right)\left(\frac{f_y}{f'_c}\right) = \left(\frac{7.85}{(24)(14)}\right)\left(\frac{60}{4}\right) = 0.35$$

From Parme's charts: Fig. 11

$$\therefore \beta = 0.66$$

$$\therefore N = \frac{\log 0.50}{\log 0.66} = 1.67$$

Eqn. (18) but use $\phi = 0.9$ since pure bending

$$\left[\frac{\left(\frac{200}{0.9}\right)}{379.7}\right]^{1.67} + \left[\frac{\left(\frac{85}{0.9}\right)}{190.6}\right]^{1.67} = 0.718 < 1.0$$

$\therefore$ Section okay for pure biaxial bending.

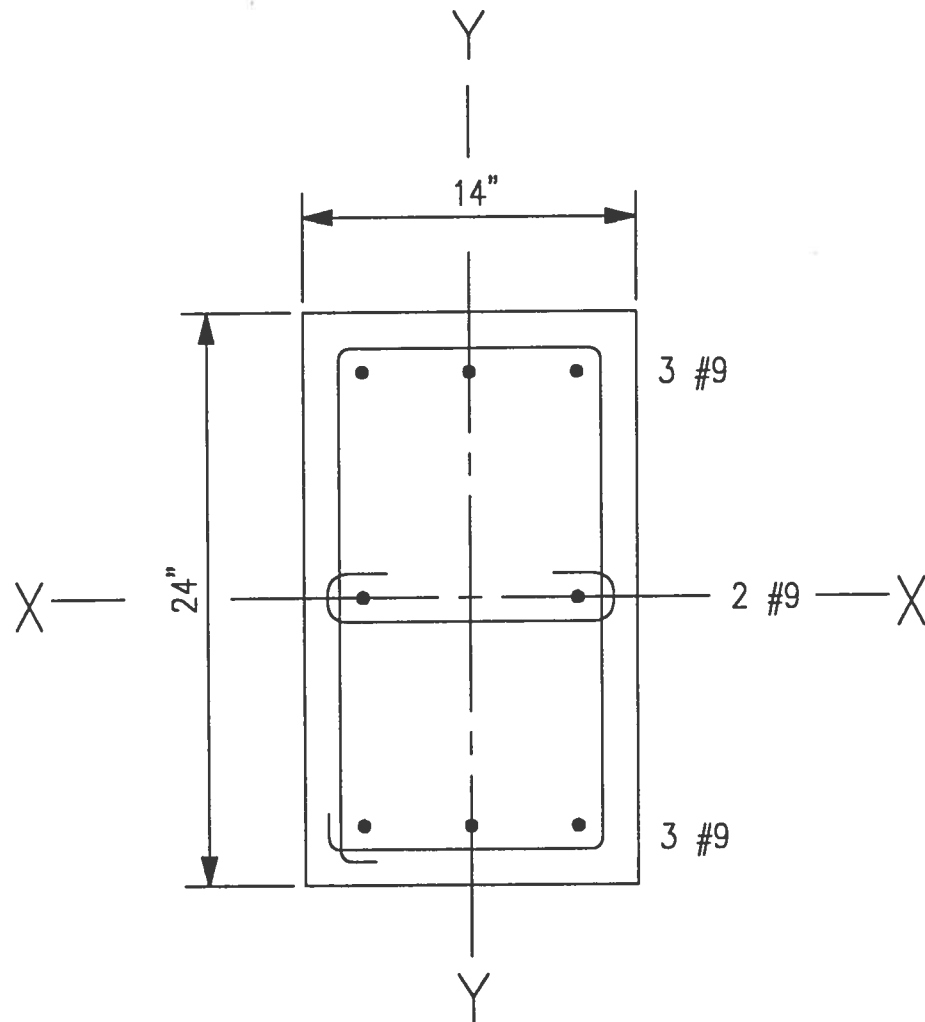
Now add steel for T_u :

$$\text{Eqn. (19)} \quad A_{\text{ST}} = A_{\text{st}} + \frac{T_u}{\phi f_y} = 7.85 + \frac{150}{0.9(60)} = 10.63 \text{ in}^2$$

$\therefore$ Use section as shown on Sketch 1

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

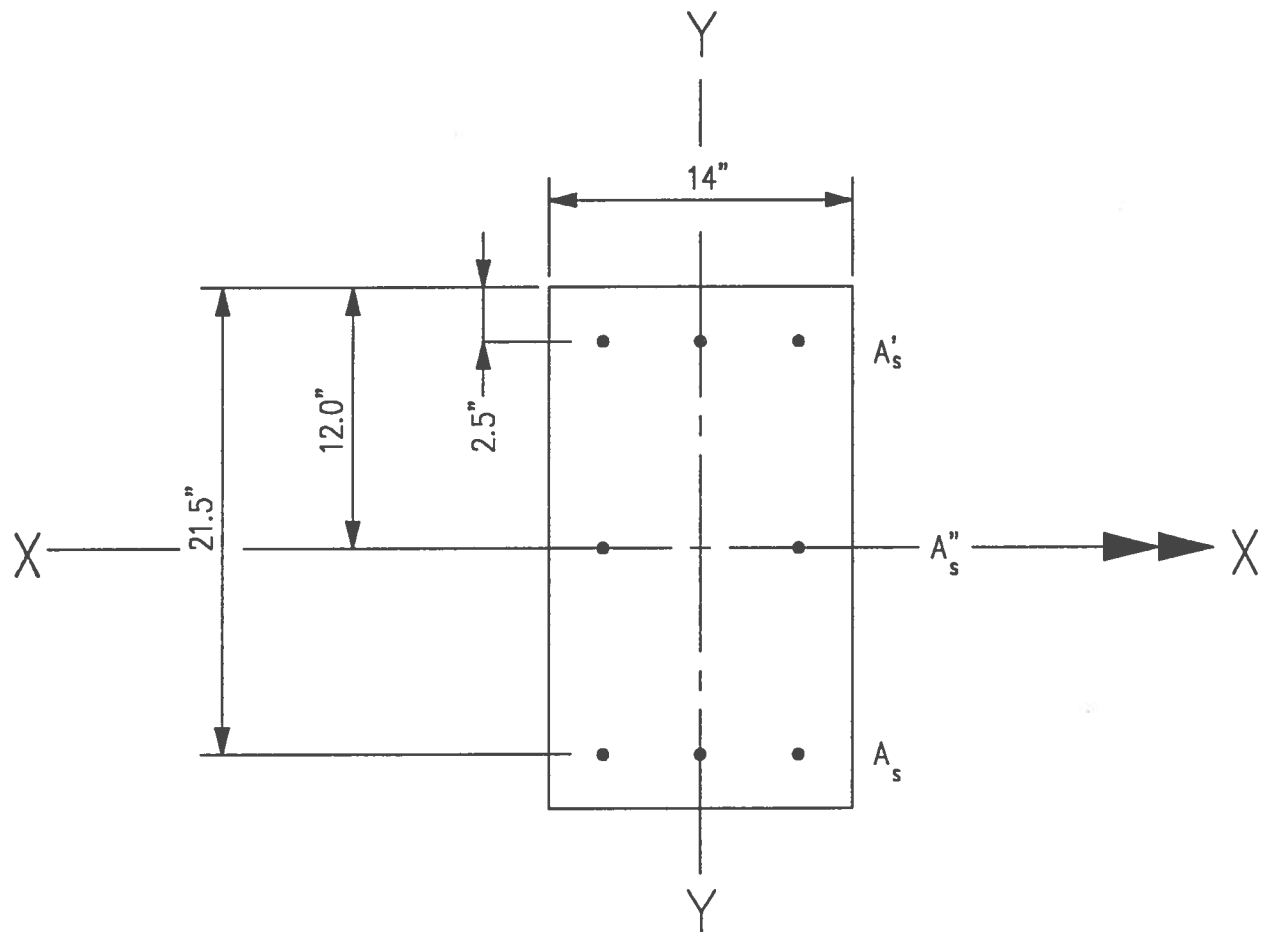


10 #9 WITH #3 @ 14" O/C (SETS OF 2)

SKETCH 1

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS



BENDING ABOUT THE X-AXIS

SKETCH 2

$$A'_s = 3 \#8 = 3 \text{ in}^2$$

$$d' = 2.5''$$

$$A''_s = 2 \#8 = 2 \text{ in}^2$$

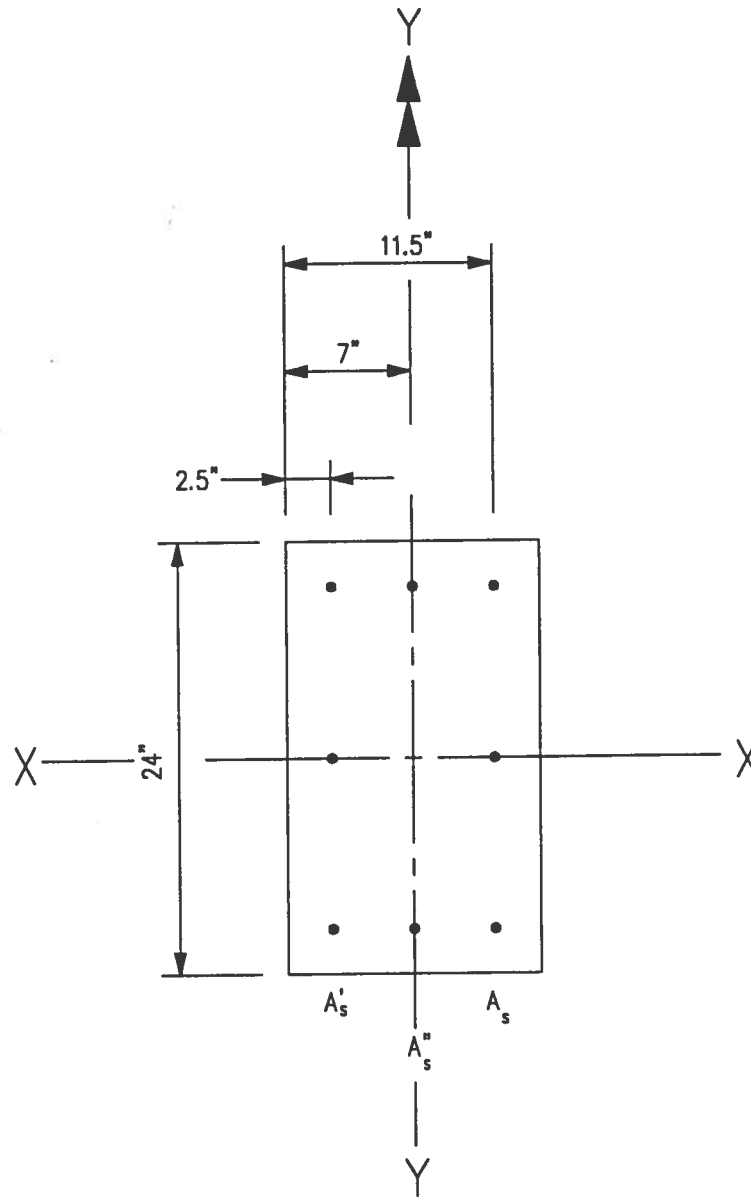
$$d'' = 12''$$

$$A_s = 3 \#8 = 3 \text{ in}^2$$

$$d = 21.5''$$

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS



BENDING ABOUT THE Y-AXIS

SKETCH 3

$$A'_s = 3 \text{ in}^2 \quad d' = 2.5''$$

$$A''_s = 2 \text{ in}^2 \quad d'' = 7.0''$$

$$A_s = 3 \text{ in}^2 \quad d = 11.5''$$

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

GIVEN: Sway frames in transverse direction
Braced frames in longitudinal direction

Top of columns: $M_{bx} = 136 \text{ K}$
 $M_{sx} = 50 \text{ K}$ Sway frames
 $M_{by} = 56 \text{ K}$
 $M_{sy} = 0$ Braced frames

Bottom of column: $M_{bx} = 105 \text{ K}$
 $M_{sx} = 60 \text{ K}$ Sway frames
 $M_{by} = 78 \text{ K}$
 $M_{sy} = 0$ Braced frames

where M_g is due to gravity loads
 M_w is due to lateral wind loads

Unsupported height of column, $l_u = 14'$

Effective length factor in the sway direction $k = 1.5$ (calculated from the frame properties).

The column bends in single curvature about each axis.

Ratio of maximum sustained factored gravity load to maximum total factored load on the column, $\beta_d = 0.5$.

Factored axial load occurring with moments, $P_u = 500 \text{ K}$

$f'_c = 4 \text{ ksi}$
 $f_y = 60 \text{ ksi}$

Design an adequate column section for the above loads and conditions.

SOLUTION - Using Spread sheet-type calculation

(a) To estimate a trial section

At this stage, without a preliminary section, we do not know if slenderness should be considered. Again the problem is to select a reasonable section for detailed analysis. Proceed as in Sample Design No. 1 to get a preliminary section. Check if slenderness is a consideration, then modify the preliminary section to get a trial section for further detailed analysis. This procedure is demonstrated as follows:

Top of column:

$$M_{ux} = 136 + 50 = 186 \text{ K}$$

$$M_{uy} = 56 + 0 = 56 \text{ K}$$

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SAMPLE DESIGN 4: BIAxIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

Since h/b is not given, use $h/b = \frac{186}{56} = 3.32$ but not greater than 2 / 1

∴ Use $h/b = 2$

Eqn. (14) $M_{\text{uox}} = 1.6 \sqrt{(186)^{1.6} + (2 \times 56)^{1.6}} = 234 \text{ K}$

Eqn. (15) $h^2 b = \frac{93(234 \times 12)}{60} = 4352.4$

$h/b = 2$

$h^3 = (4352.4)(2)$

∴ $h = 20.6''$ and $b = 10.3''$

Bottom of column:

$M_{\text{ux}} = 105 + 60 = 165 \text{ K}$

$M_{\text{uy}} = 78 + 0 = 78 \text{ K}$

Repeating calculation we get:

$h/b = \frac{165}{78} = 2.1$ but not greater than 2 / 1

∴ use $h/b = 2$

$h/b = 2$

$M_{\text{uox}} = 247.6 \text{ K}$

$h = 21.0'', b = 10.5''$

Choose the larger 'h' from the top and bottom of the column:

∴ $h = 21''$

Choose the larger 'b' from the top and bottom of the column:

∴ $b = 10.5''$

In this sample design, both the larger values of h and b occur at the bottom of the column, but this need not be so.

At this point have the spread-sheet print out:

$(h, b) = (21.0, 10.5)$

Round off the values, thus:

$(h, b) = (22, 11)$

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

and check for slenderness using ACI sections 10.11.4.1 and 10.11.4.2.

Transverse sway frame

$$\frac{kl_u}{r} = \frac{(1.5)(14 \times 12)}{(0.3 \times 22)} = 40 > 22 \text{ but } < 100 \dots \text{ACI Section 10.11.4.3}$$

$\therefore$ Slenderness occurs for sway frame.

Longitudinal braced frame

$$\frac{kl_u}{r} = \frac{(1.0)(14)(12)}{.3(11)} = 50.9$$

$$34 - 12 \frac{M_{1b}}{M_{2b}} = 34 - 12 \left(\frac{56}{78} \right) = 25.4 \dots \text{ACI Section 10.11.4.1}$$

$$\therefore \frac{kl_u}{r} = 50.9 > 25.4 \text{ but } < 100 \dots \text{ACI Section 10.11.4.3}$$

$\therefore$ Slenderness occurs for braced frame.

Slenderness must be considered in both directions, and trial section must be re-evaluated.

Assume $\delta M_{uox} = 1.5(247.6) = 371.4$ K, where δ = magnification factor assumed at this stage (see Narrative)

M_{uox} = larger equivalent moment at top and bottom of column

Re-estimate h: eqn. (15)

$$h^2 b = \frac{93 M_{uox}}{f_y}, \text{ replacing } M_{uox} \text{ with } \delta M_{uox}$$

$$\therefore h^2 = \frac{93(371.4)(12)}{60} = 6908.0$$

$h / b = 2$ as before:

$$\therefore h = 24.0", \quad b = 12.0"$$

$$\text{Eqn. (16) } A_{st} = 0.03 h b = 0.03 (24)(12) = 8.64 \text{ in}^2$$

At this point have the spread-sheet print out:

$$(h, b, A_{st}) = (24, 12, 8.64)$$

Note: Carry the decimal places, ie. no rounding off.

The actual section chosen for further detailed analysis is:

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

$$(h, b, A_s) = (24, 14, 8.0)$$

shown in Sketch 1, to satisfy the minimum h and b, and provide good spacing for the reinforcing steel.

(b) Now to analyze the section for biaxial bending and slenderness:

ACI Section 10.11.5.2

$$EI = \frac{E_c \frac{I_g}{5} + E_s I_{se}}{1 + \beta_d}$$

$$\text{where } E_c = 57000 \sqrt{f'_c} = 57000 \sqrt{4000} = 3.6 \times 10^6 \text{ psi} = 3600 \text{ ksi}$$

$$E_s = 29000 \text{ ksi}$$

About X-axis

$$E_c \frac{I_g}{5} = (3600) \frac{\left(\frac{(14)(24)^3}{12}\right)}{5} = 11.61 \times 10^6 \text{ k in}^2$$

$$E_s I_{se} = (29000)[3 \text{ in}^2 (9.5)^2] \times 2 = 15.70 \times 10^6 \text{ k in}^2$$

To find δ_{bx} for gravity loads:

$$\beta_d = 0.5$$

$$(EI)_x = \frac{11.61 \times 10^6 + 15.70 \times 10^6}{1 + 0.5} = 18.21 \times 10^6 \text{ k in}^2$$

$$\text{ACI Eqn. 10-9 } P_{cx} = \frac{\pi^2 EI}{(kl_u)^2} = \frac{\pi^2 (18.21 \times 10^6)}{[1.0(14 \times 12)]^2} = 6368 \text{ k}$$

$$\text{ACI Eqn. 10-12 } C_{mx} = 0.6 + 0.4 \frac{M_{1x}}{M_{2x}} = 0.6 + 0.4 \left(\frac{105}{136}\right) = 0.91$$

$$\text{ACI Eqn. 10-7 } \delta_{bx} = \frac{C_{mx}}{1 - \frac{P_u}{\phi P_c}} = \frac{0.91}{1 - \frac{500}{0.7(6368)}} = 1.03$$

To find δ_{sx} for lateral loads

$$\beta_d = 0 \text{ since lateral loads are temporary (wind)}$$

$$k = 1.5 \text{ for sway condition}$$

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

$$(EI)_x = \frac{11.61 \times 10^6 + 15.70 \times 10^6}{1 + 0} = 27.31 \times 10^6 \text{ k in}^2$$

$$P_{cx} = \frac{\pi^2 (27.31 \times 10^6)}{[1.5(14 \times 12)]^2} = 4244 \text{ k}$$

$$\delta_{sx} = \frac{1}{1 - \frac{\sum P_u}{\phi \sum P_c}}$$

For this example, assume 2 columns in each frame, each loaded with 500K.

$$\therefore \delta_{sx} = \frac{1}{1 - \frac{500 + 500}{0.7(4244 + 4244)}} = 1.20$$

About Y-axis

$$E_c \frac{I_y}{5} = (3600) \frac{\left(\frac{(24)(14)^3}{12}\right)}{5} = 3.95 \times 10^6 \text{ k in}^2$$

$$E_s I_{sc} = (29000)[(3 \text{ in}^2)(4.5)^2] \times 2 = 3.52 \times 10^6 \text{ k in}^2$$

To find δ_{by} for gravity loads:

$$\beta_d = 0.5, \quad k = 1.0$$

$$(EI)_y = \frac{3.95 \times 10^6 + 3.52 \times 10^6}{1 + 0.5} = 4.98 \times 10^6 \text{ k in}^2$$

$$P_{cy} = \frac{\pi^2 EI}{(kl_u)^2} = \frac{\pi^2 (4.98 \times 10^6)}{[(1.0)(14 \times 12)]^2} = 1741 \text{ k}$$

$$C_{my} = 0.6 + 0.4 \frac{M_{1b}}{M_{2b}} = 0.6 + 0.4 \left(\frac{56}{78}\right) = 0.89$$

$$\delta_{by} = \frac{C_{my}}{1 - \frac{P_u}{\phi P_c}} = \frac{0.89}{1 - \frac{500}{0.7(1741)}} = 1.51$$

To find δ_{sy} for lateral loads

Problem states that frames are braced in long direction, therefore

$$\delta = 1.0$$

Check Minimum eccentricity:

Sections 10.11.5.4 and 10.11.5.5

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SAMPLE DESIGN 4: BIAxIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

Note: Minimum eccentricity requirements now applies only to Slender columns. Where slenderness is not a consideration, the Code now allows 'zero' minimum eccentricity.

Thus, since the column is slender minimum eccentricity must be considered.

Slenderness about the X-axis

$$M_{2b} \geq P_u (0.6 + 0.03h) \\ = 500 (0.6 + 0.03(24)) = 660^{K'} = 55^{K'} < 136^{K'} \text{ given } \dots \text{OK}$$

$$M_{2s} \geq P_u (0.6 + 0.03h) = 55^{K'} \text{ as above } < 60^{K'} \text{ OK}$$

Slenderness about Y-axis

$$M_{2b} \geq 500 (0.6 + 0.03(14)) = 510^{K'} = 42.5^{K'} < 78^{K'} \text{ OK}$$

∴ Minimum eccentricity does not govern. Use computed moments to find M_{cx} and M_{cy} .

Magnified Loads due to slenderness:

Top of Columns

$$M_{cx} = \delta_{bx} M_{bx} + \delta_{sx} M_{sx} = 1.03(136) + 1.20(50) = 200^{K'}$$

$$M_{cy} = \delta_{by} M_{by} + \delta_{sy} M_{sy} = 1.51(56) + 1.0(0) = 85^{K'}$$

Bottom of Columns

$$M_{cx} = \delta_{bx} M_{bx} + \delta_{sx} M_{sx} = 1.03(105) + 1.20(60) = 180^{K'}$$

$$M_{cy} = \delta_{by} M_{by} + \delta_{sy} M_{sy} = 1.51(78) + 1.0(0) = 118^{K'}$$

$$P_u = 500^K \text{ given}$$

In this example, only the bottom section of the column will be checked, since it appears to be more critical. However, both top and bottom sections should be checked.

Bottom of Columns

$$M_{cx} = 180^{K'}$$

$$M_{cy} = 118^{K'}$$

$$P_u = 500^K$$

Find β

ACI Section 10.3.5

$$\text{ACI Eqn. 10-2 } P_{no} = 0.80 [0.85f'_c (A_g - A_{st}) + f_y A_{st}]$$

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

$$= 0.80 [0.85(4)(14 \times 24 - 8) + (60)(8)] = 1276^k$$

$$P_u / \phi = 500 / 0.7 = 714.3$$

$$\therefore \frac{(P_u / \phi)}{P_m} = \frac{714.3}{1276} = 0.56$$

$$q = \rho \frac{f_y}{f'_c} = \left(\frac{A_s}{h b} \right) \left(\frac{f_y}{f'_c} \right) = \left(\frac{8.0}{(14)(24)} \right) \left(\frac{60}{4} \right) = 0.36$$

From Parme's charts or "Look-up Tables", $\beta = 0.63$

Eqn. (17) $N = \frac{\log 0.5}{\log 0.63} = 1.5$

Find M_{nox} and M_{noy} @ $P_n = P_u / \phi = 714.3^k$

Proceed as in Sample Design 2A for Bending About the X-Axis:

At balanced condition:

$$c_b = \frac{0.003}{0.003 + 0.00207} (21.5) = 12.72"$$

$$C' = 180.0^k$$

$$C = 514.7^k$$

$$C'' = 9.85^k$$

$$T = 180.0^k$$

$$\therefore \Sigma C = 704.6^k, \quad \Sigma T = 180.0^k$$

$$\therefore P = 524.6^k < P_u / \phi = 714.3^k$$

$$\therefore c > c_b$$

Try $c = 16"$

$$C' = 180.0^k$$

$$C = 647.4^k$$

$$C'' = 43.5^k$$

$$T = 180.0^k$$

$$\therefore P = 690.9^k < P_u / \phi = 714.3^k$$

$$\therefore c > 16"$$

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SAMPLE DESIGN 4: BIAXIAL BENDING AND AXIAL COMPRESSION FOR SLENDER COLUMNS

Try $c = 16.5"$

$$C' = 180.0^k$$

$$C = 667.6^k$$

$$C'' = 47.5^k$$

$$T = 180.0^k$$

$$\therefore P = 715^k \cong P_u / \phi = 714.3^k$$

$$\therefore \text{Use } P = P_n = 714.3^k \text{ close enough}$$

$$\therefore P_n e' = C' l'_a + C l_a + C'' l''_a$$

$$714.3 e' = (180)(21.5 - 2.5) + 667.6(21.5 - \frac{0.85(16.5)}{2}) + 47.5(21.5 - 12)$$

$$e' = 18.96" \quad \therefore e_{oy} = 18.96 - 21.5 + \frac{24}{2} = 9.46"$$

$$\therefore M_{nox} = \frac{(714.3)(9.46")}{12} = 563^k$$

For Bending About the Y-Axis

At balanced condition

$$c_b = \frac{0.003}{0.003 + 0.00207} (11.5) = 6.81"$$

$$C' = 165.1^k$$

$$C = 472.3^k$$

$$T'' = 4.85^k$$

$$T = 179.7^k$$

$$\therefore P = 452.9^k < P_u / \phi = 714.3^k$$

$$\therefore c > c_b$$

Try $c = 8"$

$$C' = 179.4^k$$

$$C = 554.9^k$$

$$C'' = 21.8^k$$

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SAMPLE DESIGN 4: BIAxIAL BENDING AND AxIAL COMPRESSION FOR SLENDER COLUMNS

$$T = 114.2^k$$

$$\therefore P = 663.7^k < P_u / \phi = 714.3^k$$

$$\therefore c > 8"$$

$$\text{Try } c = 8.55"$$

$$C' = 180^k$$

$$C = 593.0^k$$

$$C'' = 31.5^k$$

$$T = 90.1^k$$

$$\therefore P = 714.4^k \cong P_u / \phi = 714.3^k$$

$$\therefore \text{Use } P = P_n = 714.3^k \text{ close enough}$$

$$\therefore P_n e' = C' l'_a + C l_a + C'' l''_a$$

$$e' = 9", \therefore e_{ox} = 4.5"$$

$$\therefore M_{\text{noy}} = \frac{(714.3)(4.5")}{12} = 267.6^k$$

$$\text{Eqn. (20)} \quad \left(\frac{257.1}{564}\right)^{1.5} + \left(\frac{168.6}{267.6}\right)^{1.5} = 0.808 < 1.0$$

$\therefore$ Section is adequate for bottom of column.

A similar procedure should be used to check the top of the column, for

$$M_{cx} = 200^k$$

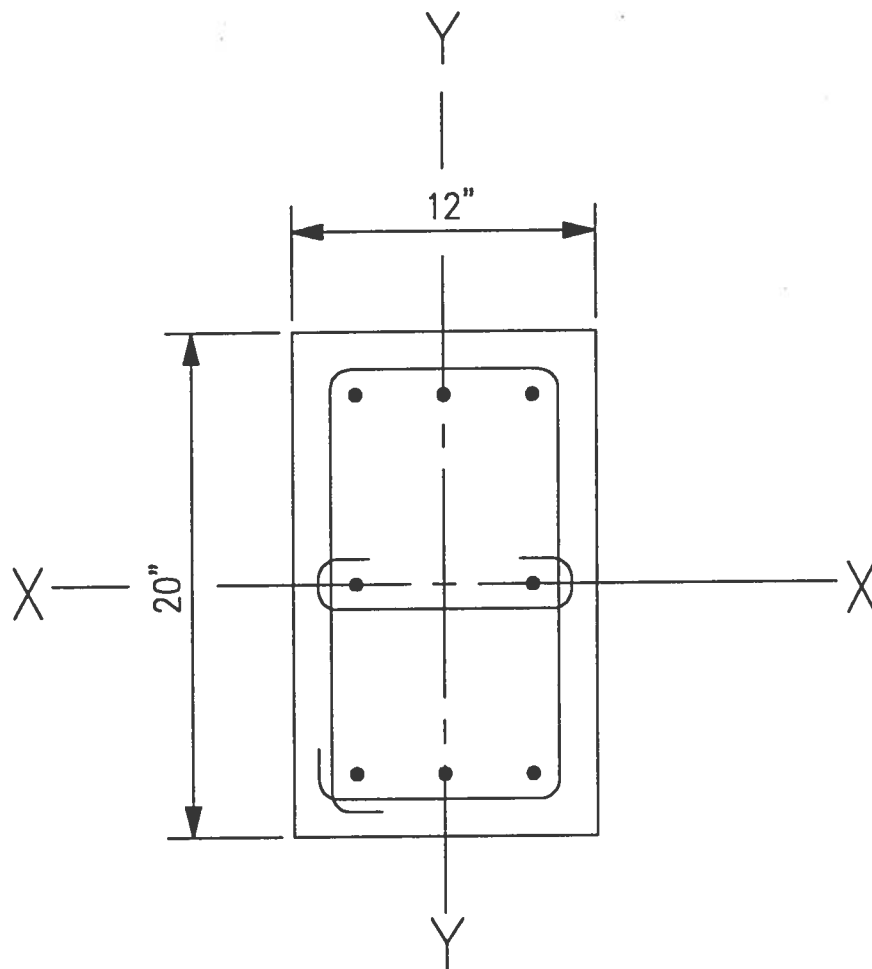
$$M_{cy} = 85^k$$

$$P_u = 500^k$$

If desired the section may also be checked using the interaction curves of ACI 340.2R-85 Design Handbook, Volume 2, as per Sample Design No. 2C or 2D.

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SAMPLE DESIGN 5: UNIAXIAL BENDING AND AXIAL COMPRESSION

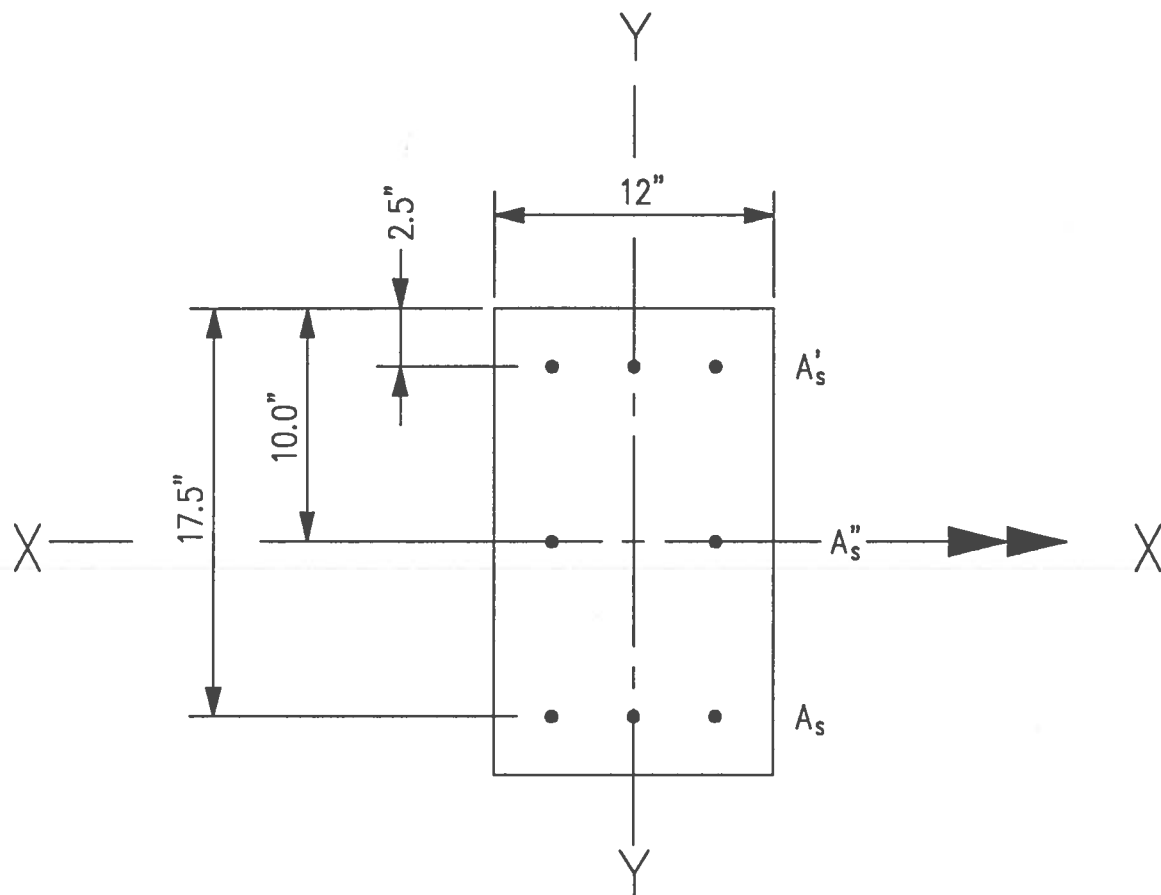


8 #8 WITH #3 TIES @ 12" O/C (SETS OF 2)

SKETCH 1

FLUOR DANIEL

SAMPLE DESIGN 5: UNIAXIAL BENDING AND AXIAL COMPRESSION



BENDING ABOUT THE X-AXIS

SKETCH 2

$$A'_s = 2.36 \text{ in}^2$$

$$d' = 2.5''$$

$$A''_s = 1.57 \text{ in}^2$$

$$d'' = 10''$$

$$A_s = 2.36 \text{ in}^2$$

$$d = 17.5''$$

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$$M_{ux} = 200 \text{ k}$$

$$P_u = 100 \text{ k}$$

$$f'_c = 4 \text{ ksi}$$

$$f_y = 60 \text{ ksi}$$

Short Column . . . ie. No slenderness.

5A: Spread-sheet Type calculation

Estimate trial section using Eqn. 15:

$$\text{Eqn. (15)} \quad h^2b = \frac{93 M_{ux}}{f_y} \text{ except that } M_{uox} = M_{ux} = 200 \text{ k} \quad \therefore h^2b = \frac{93(200 \times 12)}{60} = 3720$$

Since h/b is not given, assume $h/b = 2.0$

$$h^3 = 3720 \times 2$$

$$\therefore h = 19.5", \quad b = 9.8"$$

Round off to (h, b) to $(20", 10")$.

$$\text{Eqn. (16)} \quad A_{st} = (0.03)(20)(10) = 6 \text{ in}^2$$

Increase b to 12" for practical fit of reinforcement.

See ACI-HB Reinforcement 2, dimension " b_2 ".

$\therefore$ Use section on sketch 1 as trial section for further analysis

$$\text{Use compatibility and equilibrium equations (1) and (8), to find } P_n \text{ at } e_y = \frac{M_{ux}}{P_u} = \frac{200 \times 12}{100} = 24"$$

At balanced condition

$$\text{Eqn. (8)} \quad c_b = \left(\frac{0.003}{0.003 + 0.00207} \right) (17.5) = 10.36"$$

Repeating calculation similar to Sample Design No. 2A:

$$C' = 141.6 \text{ k}$$

$$C = 359.3 \text{ k}$$

$$C'' = 4.7 \text{ k}$$

$$T = 141.6 \text{ k}$$

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$$\text{Eqn. (6)} \quad P + 141.6 = 141.6 + 4.7 + 359.3 = 364.0^k$$

$$\text{Eqn. (5)} \quad l'_a = 17.5 - 2.5 = 15"$$

$$l''_a = 17.5 - 10 = 7.5"$$

$$l_a = 17.5 - \frac{0.85(10.36)}{2} = 13.1"$$

$$\text{Eqn. (7)} \quad 364 e' = (141.6)(15) + (359.3)(13.1) + (4.7)(7.5)$$

$$e' = 18.86"$$

$$\text{Fig 1(b)} \quad \therefore e = 18.86 - 17.5 + 20/2 = 11.36"$$

$$\text{ie. } e_y = 24" > e = 11.36"$$

$$\therefore c < c_b = 10.36"$$

$$\text{Try } c = 7"$$

$$C' = 132.0^k$$

$$C = 242.8^k$$

$$T'' = 58.5^k$$

$$T = 141.6^k$$

$$\therefore P = 174.7^k$$

$$e' = 29.0" \text{ and } e = 21.5"$$

$$\therefore c < 7"$$

$$\text{Try } c = 6.5"$$

$$C' = 126.4^k$$

$$C = 225.4^k$$

$$T'' = 73.5^k$$

$$T = 141.6^k$$

$$\therefore P = 136.7^k$$

$$e' = 34.1" \text{ and } e = 26.6"$$

$$\therefore c > 6.5"$$

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$$\text{Try } c = 6.7''$$

$$C' = 128.7^k$$

$$C = 232.4^k$$

$$T'' = 67.3^k$$

$$T = 141.6^k$$

$$\therefore P = 152.2^k$$

$$e' = 31.7'' \text{ and } e = 24.2'' \cong e_y = 24''$$

$$\therefore \text{Required } c = 6.7''$$

$$P = P_n = 152.2^k \text{ and } M_{nx} = 152.2 \left(\frac{24''}{12} \right) = 304.4^k$$

$$\therefore \phi P_n = 0.7 (152.2^k) = 106.5^k$$

$$\phi M_{nx} = 0.7 (304.4) = 213.1^k$$

$$\therefore \text{Section is adequate since } \phi P_n = 106.5^k > P_u = 100^k \text{ at } e_y = 24''$$

5B: Using ACI - Handbook

Use same trial section as developed in 5A . . . see sketch 1

$$\gamma = \frac{20 - 2(1.5)}{20} = 0.75$$

$\therefore$ Use ACI-HB Columns 7.4.3

$$\frac{\phi P_n}{A_g} = \frac{P_u}{A_g} = \frac{100}{(20 \times 12)} = 0.417$$

$$\frac{\phi M_{nx}}{A_g h} = \frac{M_u}{A_g h} = \frac{200 \times 12}{(20)(12)(20)} = 0.50$$

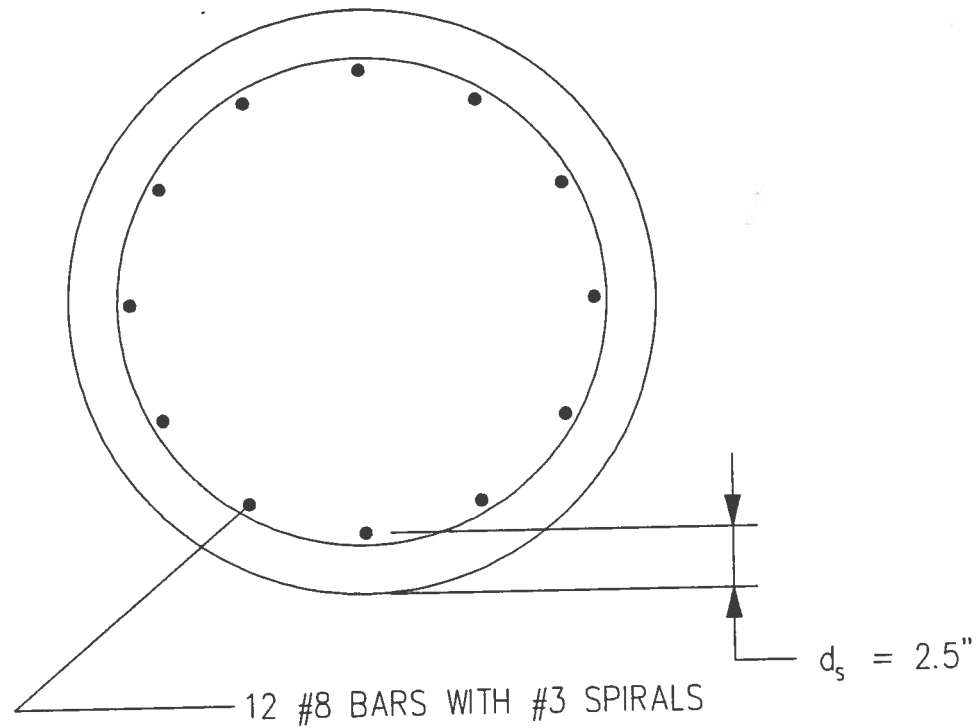
$$\therefore \text{From chart } \rho_g = 0.025 = 0.025 (20 \times 2) = 6 \text{ in}^2$$

$$\rho_g \text{ provided} = 8 \text{ \#8 bars} = 6.28 \text{ in}^2$$

$\therefore$ Section adequate

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SAMPLE DESIGN 6: CIRCULAR COLUMN



SKETCH 1

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SAMPLE DESIGN 6: CIRCULAR COLUMN

$$M_u = 200 \text{ k'}$$

$$P_u = 100 \text{ k}$$

$$f'_c = 4 \text{ ksi}$$

$$f_y = 60 \text{ ksi}$$

Short column . . . ie. No slenderness

There is no direct method for estimating the trial section, similar to the one for rectangular sections. However, try the following procedure:

$$\text{Eqn. (15)} \quad h^2 b = \frac{93 M_{uox}}{f_y} = \frac{93(200 \times 12)}{60} = 3720$$

$$\text{Use } h/b = 1, \therefore h^3 = 3720, \quad h = 15.5" \quad \text{ie. square columns of side } h = 15.5"$$

Equivalent circular column:

$$\pi r^2 = 15.5 \times 15.5$$

$$\therefore r = 8.74"$$

$$\text{Diameter} = 8.74 \times 2 = 17.5"$$

Round off to 18" diameter column.

See sketch 1.

$$\frac{\phi P_n}{A_g} = \frac{P_u}{A_g} = \frac{100}{\pi(9)^2} = 0.393$$

$$\frac{\phi M_n}{A_g h} = \frac{M_u}{A_g h} = \frac{200 \times 12}{[\pi(9)^2](18)} = 0.524$$

$$\gamma = \frac{18 - 2(2.5)}{18} = 0.72$$

$\therefore$ Use ACI-HB Columns 7.22.2 and 7.22.3 and interpolate.

$$\text{Columns 7.22.2 } \gamma = 0.60, \quad \rho_g = 0.045$$

$$\text{Columns 7.22.3 } \gamma = 0.75, \quad \rho_g = 0.033$$

$$\therefore \rho_g \text{ required} = 0.033 + \frac{(0.75 - 0.72)}{(0.75 - 0.60)} (0.045 - 0.033) = 0.0354 = (0.0354)\pi(9)^2 = 9 \text{ n}^2 = 12 \text{ \#8 bars.}$$

Note: The 'trial section' procedure appeared to give a reasonable section for further analysis. The author would appreciate feedback on its consistency to provide a good trial section, based on the many designs of columns to be carried out by the use of this practice. Of course, other considerations, such as architectural, may dictate the choice of the column dimensions.